香港中文大學
The Chinese University of Hong Kong

ROSE5710 Fundamental Robotics and Systems

Lecture Set 1: What is Robotics?

Darwin Lau

2026-27

Lecture Summary

Aims of the lecture

- Introduction on the history and development of robots
- Present the different types and applications of robots
- Discuss the different challenges involved and newest trends

References

- Mostly from videos available on YouTube

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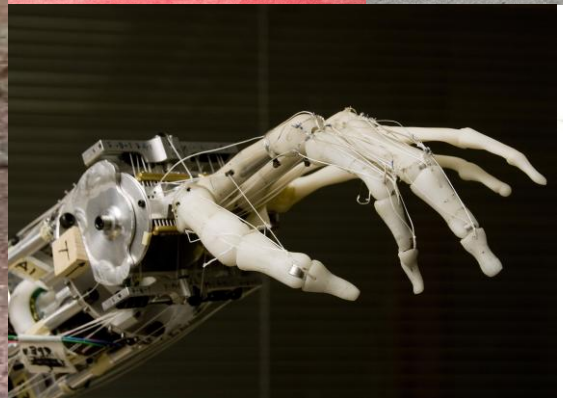
What is a Robot?

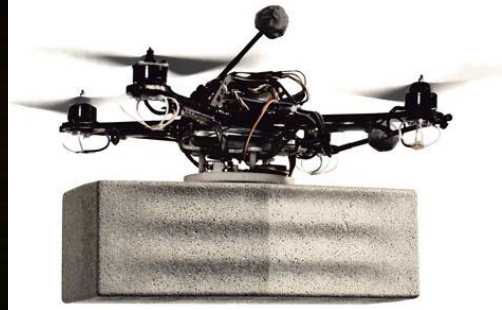












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Discussion Exercise: Which is a Robot?













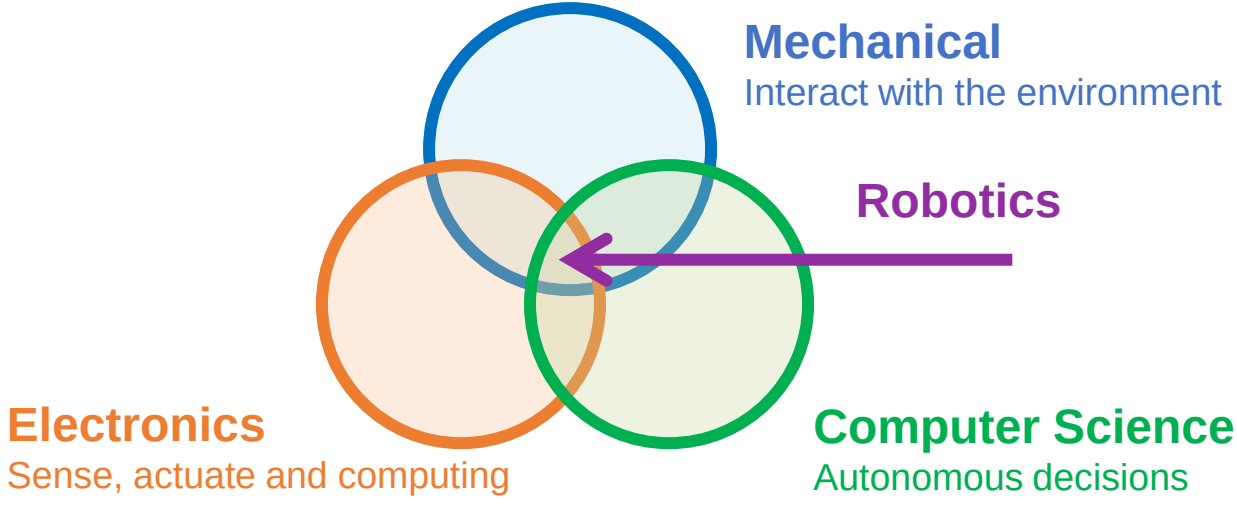




Definition of a Robot

In my opinion, no strict definition, but it is well accepted that a modern robot should have:

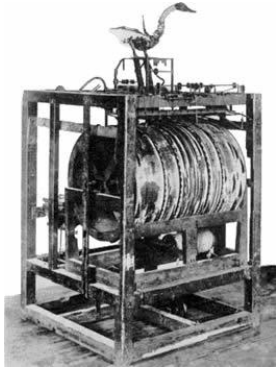
- **Mechanical elements** to interact with the environment
- **Electronics** to sense, actuate and process information
- **Software** to enable the system to be autonomous



Robotics Timeline: Fiction Era

Milestones, the **fiction era**…

- 1738: Jacques de Vaucanson “moving automata” duck
- 1893: Prof. George Moore’s “mechanical man”
- 1921: Writer Karel Čapek introduced the word “robot”
- 1942: Isaac Asimov Three Laws of Robotics
- 1956: Devol and Engelberger founded Unimation



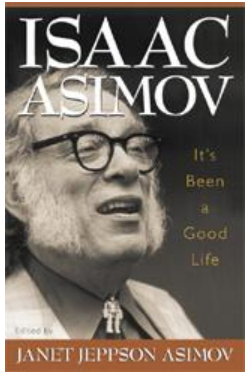
Wikipedia Commons



The New York Times



Wikipedia Commons



Book: It’s Been a Good Life



Los Angeles Times

First Time Mass Production Is Automated

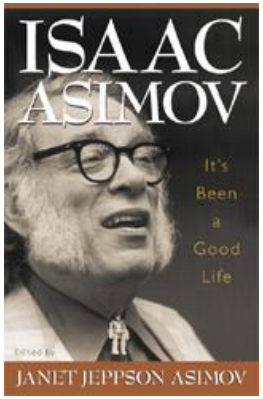
In 1804, a French inventor named Joseph-Marie Jacquard unveiled what would become the widely adopted “Jacquard Loom.”

- translating patterns from punch cards into commands that determined the lifting and lowering of threads
- increasing the speed at which complex patterns could be woven by more than twenty times, from one inch to two feet a day.



13 Milestones in the History of Robotics | Aventine

Robotics Timeline: Fiction Era



Isaac Asimov **Three Law of Robotics**

Isaac Asimov is credited with the popularization of robotics in a collection of short stories published between 1938 and 1942. Asimov is best known for his three laws governing robot behavior.

1. A robot may not injure a human being or through inaction allow a human to come to harm.
2. A robot must obey orders given it by humans except when doing so conflicts with the first law.
3. A robot must protect its own existence as long as this does not conflict with the first or second law.

First Machine Navigates on Its Own

In 1949, a neurophysiologist and inventor named William Grey Walter introduced a pair of battery-powered, tortoise-shaped robots

- Maneuver around objects in a room
- Guide themselves toward a source of light and find their way back to a charging station
- Consists of: 1) sensor technology, 2) a responsive feedback loop, and 3) logical reasoning

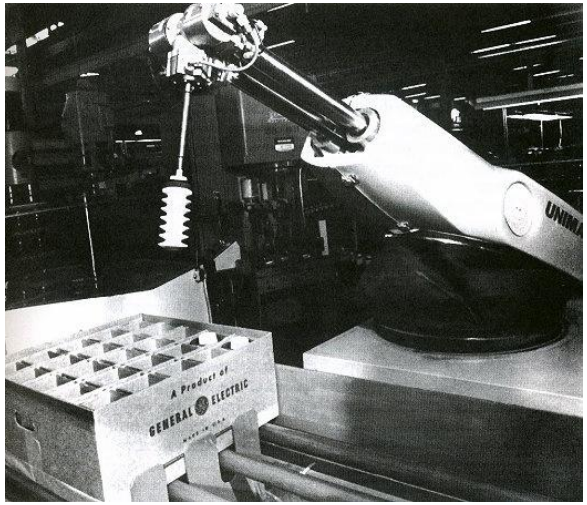


13 Milestones in the History of Robotics | Aventine

Robotics Timeline: The First Factory Robot

1961: **Unimates** first robot manipulator

- Foundation of the modern robotics industry
- “Programmable transfer machines” stored motion
- Weighed 4000 lb (1814 kg) with accuracy 0.05” (1.3mm)





The Unimate was a robot arm that weighed 4,000 pounds. It was added to the General Motors assembly line in 1961.

Robotics Timeline: The First Factory Robot

Story of UniMate: George Devol and Joseph Engelberger



Robotics Timeline: 1960s

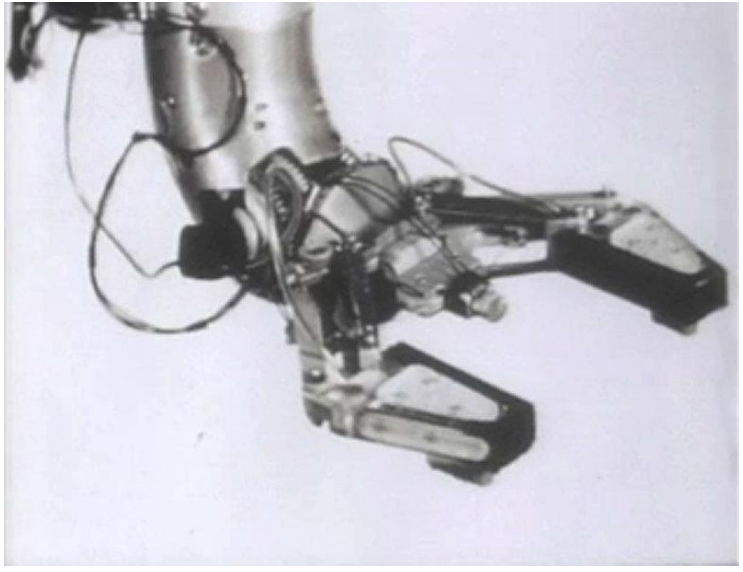
The 1960s saw the ***beginning of modern robotics…***

- 1959: Artificial Intelligence Laboratory formed at MIT
- 1961: Heinrich Ernst develops the MH-1 computer operated hand
- 1963: Artificial Intelligence Laboratory formed at Stanford
- 1966: Stanford Research Institute creates Shakey, the first mobile robot to reason about its actions
- 1969: Stanford AI Laboratory (SAIL) creates the Stanford Arm, the first computer controlled arm

Robotics Timeline: 1960s

MH-1 computer operated hand

- Controlled by the TX0 computer
- Make automated decisions



MIT MH-1 robot hand

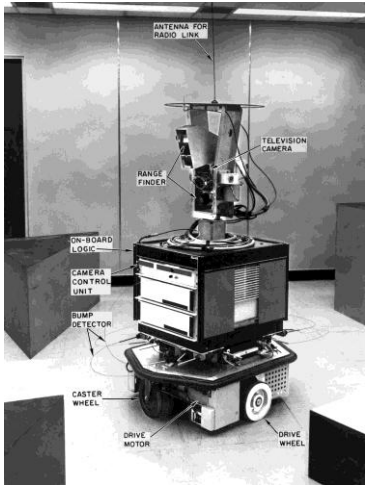


MIT TX-0 computer

Robotics Timeline: 1960s

Shakey Robot (first robot to use AI)

- Can plan and reason
- PDP-10 remote comp.
- Radio communication
- I/O via teletype





Robotics Timeline: 1960s

Stanford Arm (1969)

- Designed exclusive for computer control (calculations in real-time)
- The 6 DoF configured such that the computation of kinematic solutions were simplified



The Stanford Arm

Robotics Timeline: 1970s

The 1970s: use of robots in manufacturing and beyond…

- 1970: The Stanford Cart, line following robot (camera)
- **1971: Invention of the microprocessor**
- **1973: Kuka built its first six electric motor robot arm FAMULUS**
- 1973: Wabot-1 first anthropomorphic humanoid robot
- **1974: The first microcontroller controlled electric industrial robot IRB 6 from ABB**
- 1976: Shigeo Horise designs the Soft Gripper at Tokyo Institute of Technology
- 1977: Starwars released with R2-D2 and C-3PO
- 1979: **Prof. Hiroshi Makino created SCARA**
- 1979: Robotics Institute at CMU established

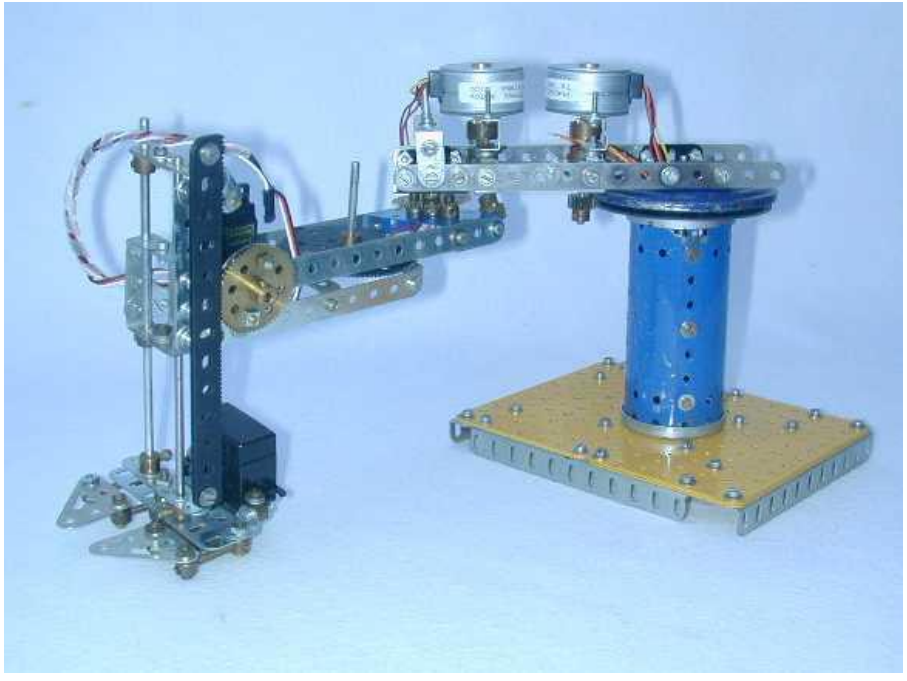
Robotics Timeline: 1970s

The soft gripper (1976)

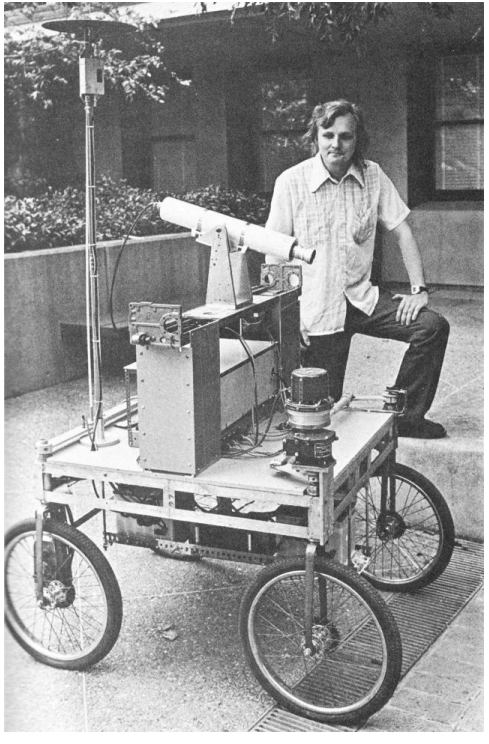


Robotics Timeline: 1970s

SCARA robot for pick and place



Robotics Timeline: 1970s



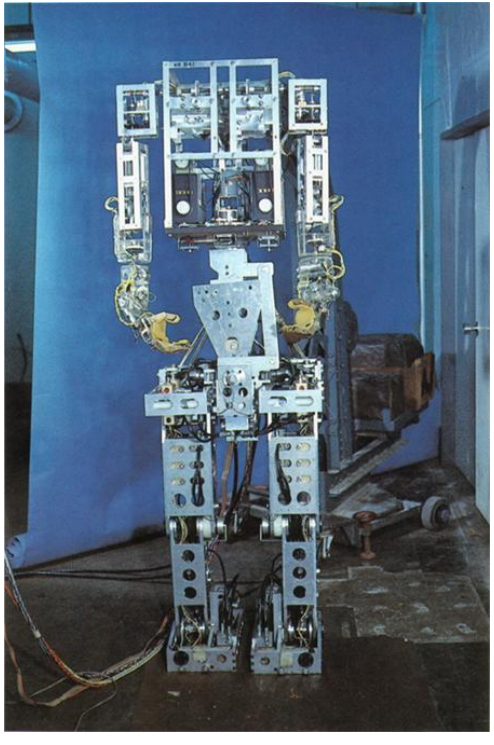
Hans Moravec with the Stanford Cart (1977).
First stereo vision for a mobile robot



The first Kuka robot: Famulus



3CPO and R2D2



WABOT-1: First fun-scale anthropomorphic robot

Robotics Timeline: 1980s

The 1980s saw advances in robot intelligence…

- 1981: Takeo Kanade created the **first direct drive arm**
- 1984: Douglas Leant started Cyc, common sense database for AI
- 1984: Wabot-2 is capable of playing the organ
- 1985: LEGO began sponsoring the MIT Media Lab
- **1986: Honda begins its humanoid R&D program (Honda E0)**
- **1988: LEGO tc Logo released**
- 1989: Chess playing programs defeated chess-masters
- **1989: Hexapod robot Genghis (4 microprocessors, 22 sensors and 12 servo motors)**

Robotics Timeline: 1980s



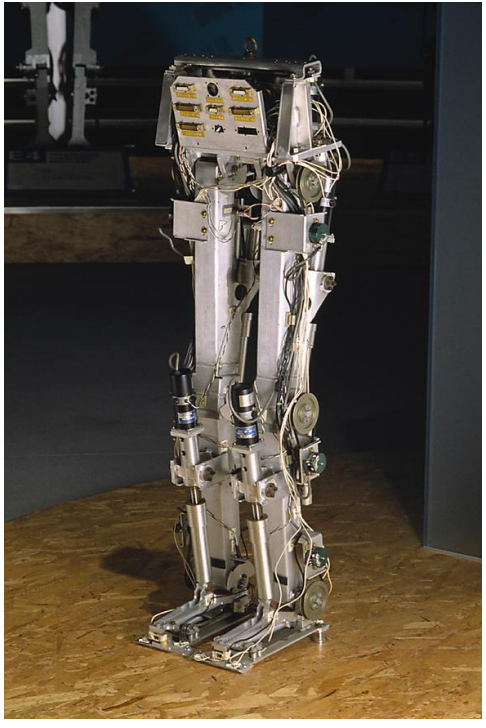
Wabot-2 playing the keyboard



Genghis robot: a six legged hexapod robot



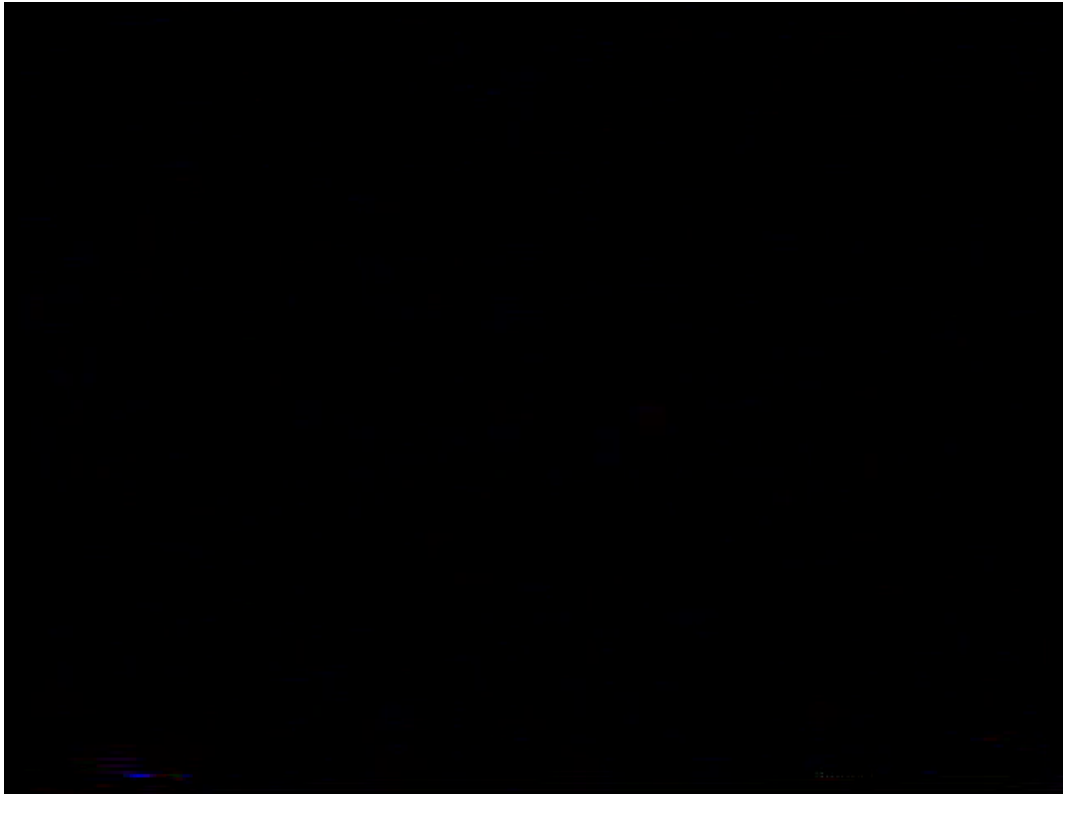
LEGO tc Logo kit



Honda E0 robot

Robotics Timeline: 1980s

LEGO: robotics in education

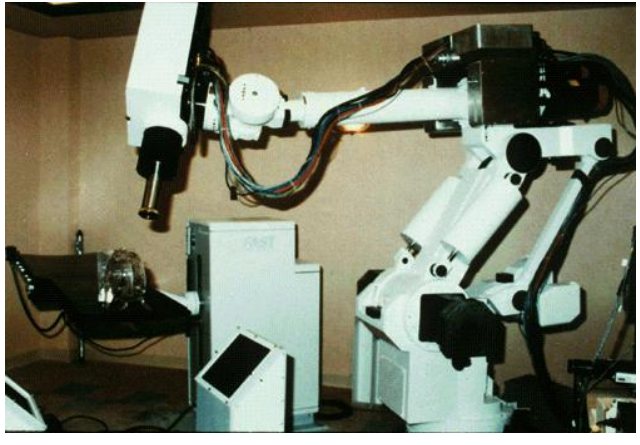


Robotics Timeline: 1990s

The 1990s saw robots getting diversified…

- **1994: Dr. John Adler invented the Cyberknife**
- 1993: Honda P1 humanoid robot
- **1996: Prof. Stuart Wilkinson invented Gastrobot**
- 1996: Dr. David Barrett built the RoboTuna (fish)
- 1996: Honda P2 humanoid robot
- 1997: Honda P3 humanoid robot
- 1998: LEGO Mindstorms first introduced
- **1999: Sony AIBO dog released**

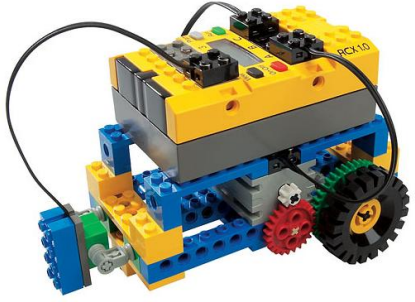
Robotics Timeline: 1990s



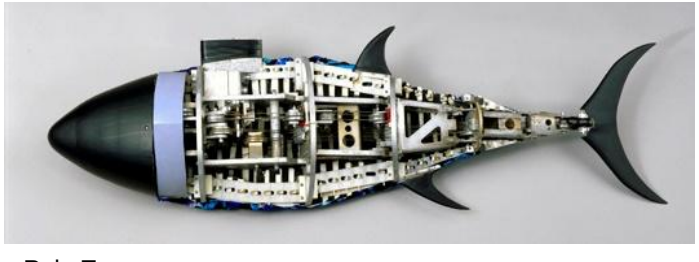
Cyberknife



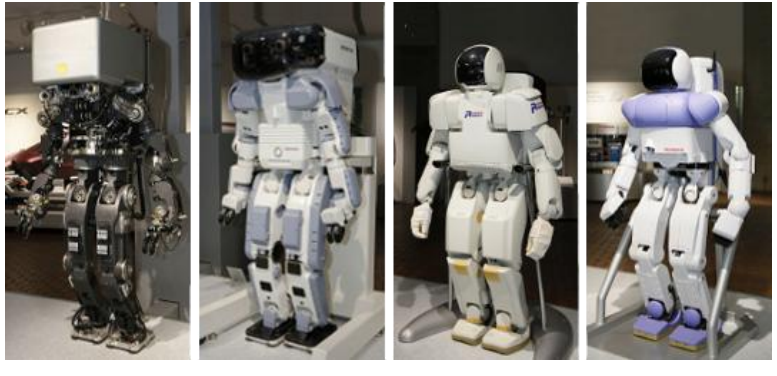
Sony AIBO



Lego Mindstorm



RoboTuna



Honda P-series, from P1 (most left), P2, P3 and P4 (most right)

Robotics Timeline: 1990s

Sony AIBO



Robotics Timeline: 2000s

The 2000s saw robots **becoming more common**

- **2000: Honda ASIMO humanoid robot**
- **2000: FDA approved the Da Vinci Surgical System**
- 2001: **FDA clears** the CyberKnife for tumour removal anywhere in the body
- 2001: iRobot PackBot searches at the US WTC
- **2002: iRobot released the Roomba vacuum cleaner**
- 2003: NASA launches the MER-A “Spirit” for Mars
- 2004: Cornell Uni revealed self replicating robots
- **2005: Arduino project started**
- 2007: Touch Bionics created i-LIMB, the first commercially available prosthetic hand

Robotics Timeline: 2000s



Honda ASIMO



iRobot Roomba



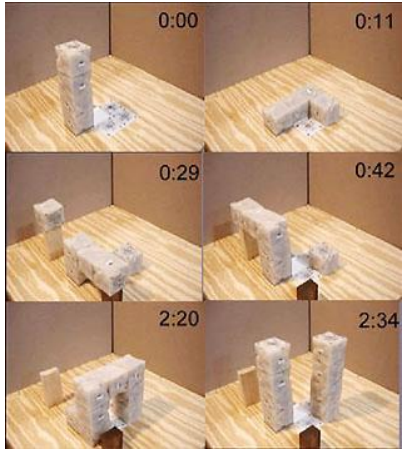
iRobot PackBot



Touch Bionics i-Limb



Da Vinci Surgery Robot

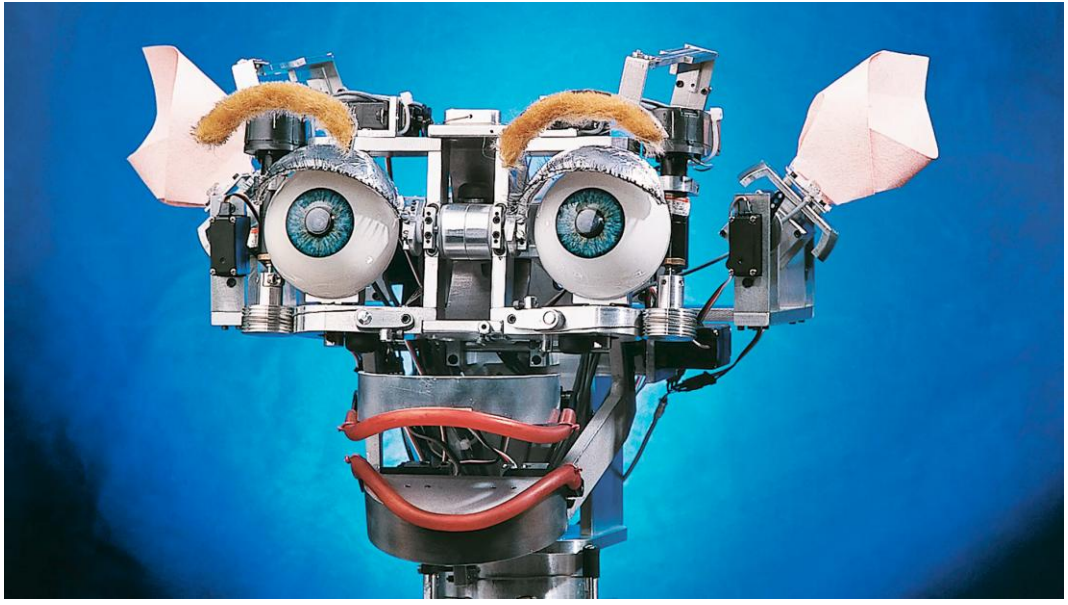


Cornell University self replicating robots

Robotics Timeline: 2000s

Social robots (2000)

- Twenty-one motors controlled an expressive pair of yellow eyebrows, red lips, pink ears and big blue eyes, allowing Kismet to express a range of emotions, from happy to bored



13 Milestones in the History of Robotics | Aventine

Robotics Timeline: 2000s

Kiva Robots (2003)

- Mick Mountz and his cofounders created Kiva for warehouse work
- Amazon bought the company for $775m in 2012



Robotics Timeline: 2000s

Big Dog by Boston Dynamics (2004)

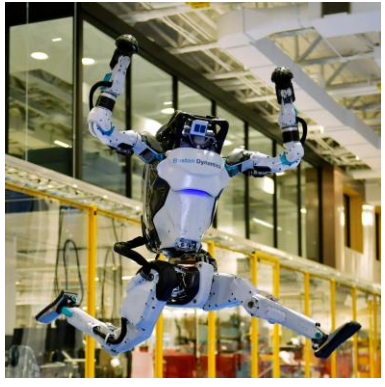


Robotics Timeline: 2010s

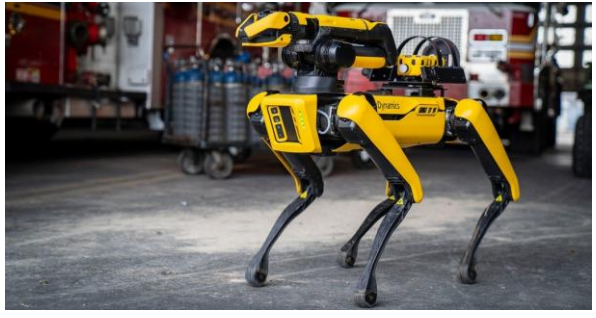
The 2010s saw advanced with the **breakthrough in AI (deep learning)**

- Robots in manufacturing
- Robots in medicine
- Robots in service
- Robots in military
- Robots in agriculture
- Robots in construction
- Robots in human assistance and collaboration
- Robots in nanotechnology

Robotics Timeline: 2010s



Boston Dynamics ATLAS



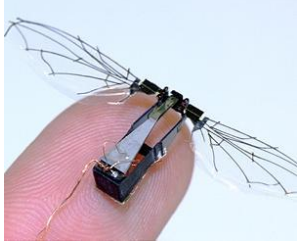
Boston Dynamics Spot



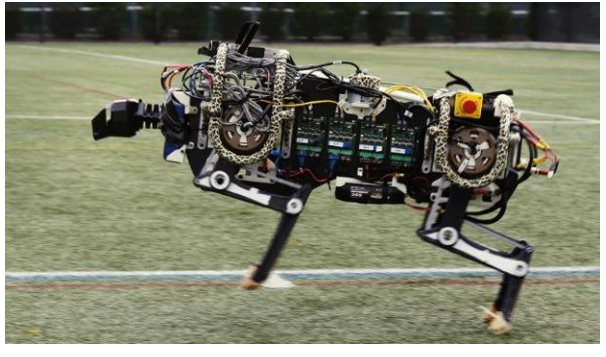
NAO robot



Aldebaran Robotics and Softbank mobile's Pepper service robot



Harvard Microfly



MIT Cheetah robot

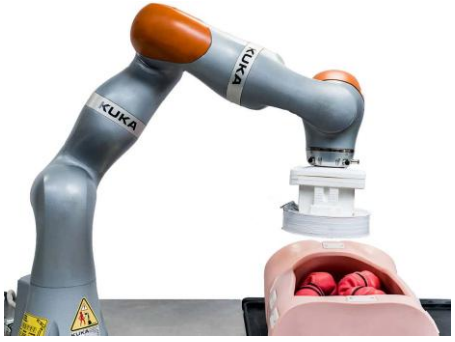
Robotics Timeline: 2020s

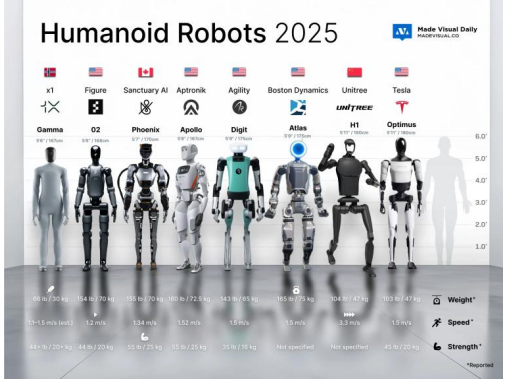
The 2020s is too exciting

- Robots with higher and higher levels of autonomy







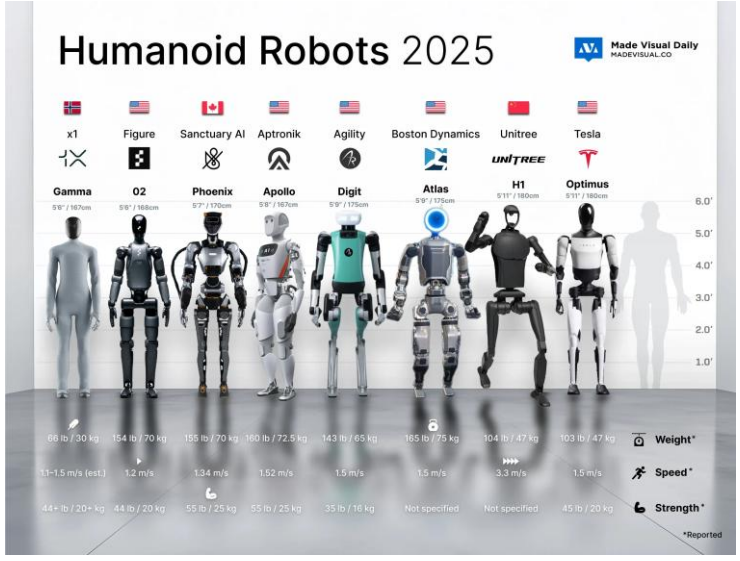






Robotics Timeline: 2020s

Are humanoids the future?





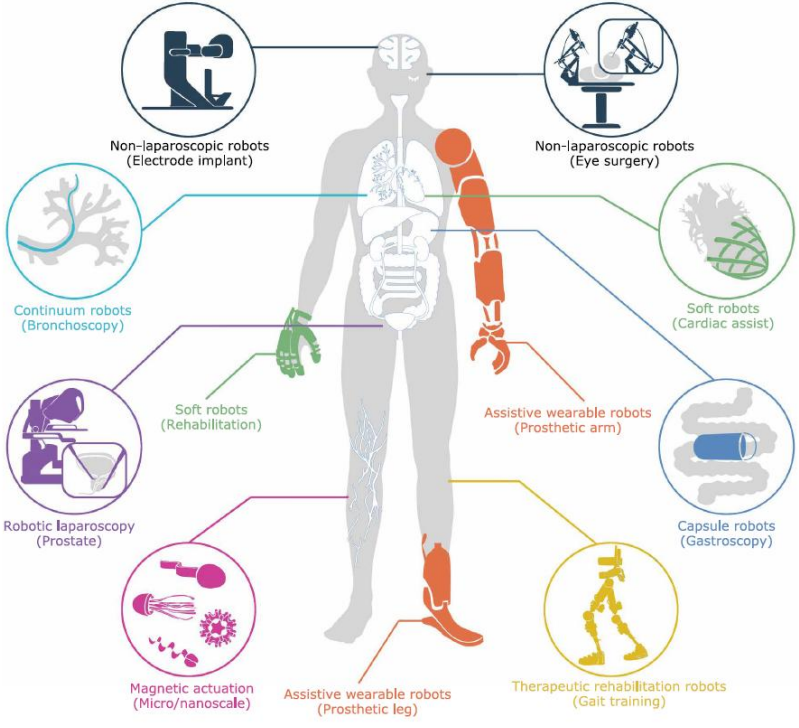
Robotics Timeline: 2020s

Cool



Medical Robotics Development and Trends

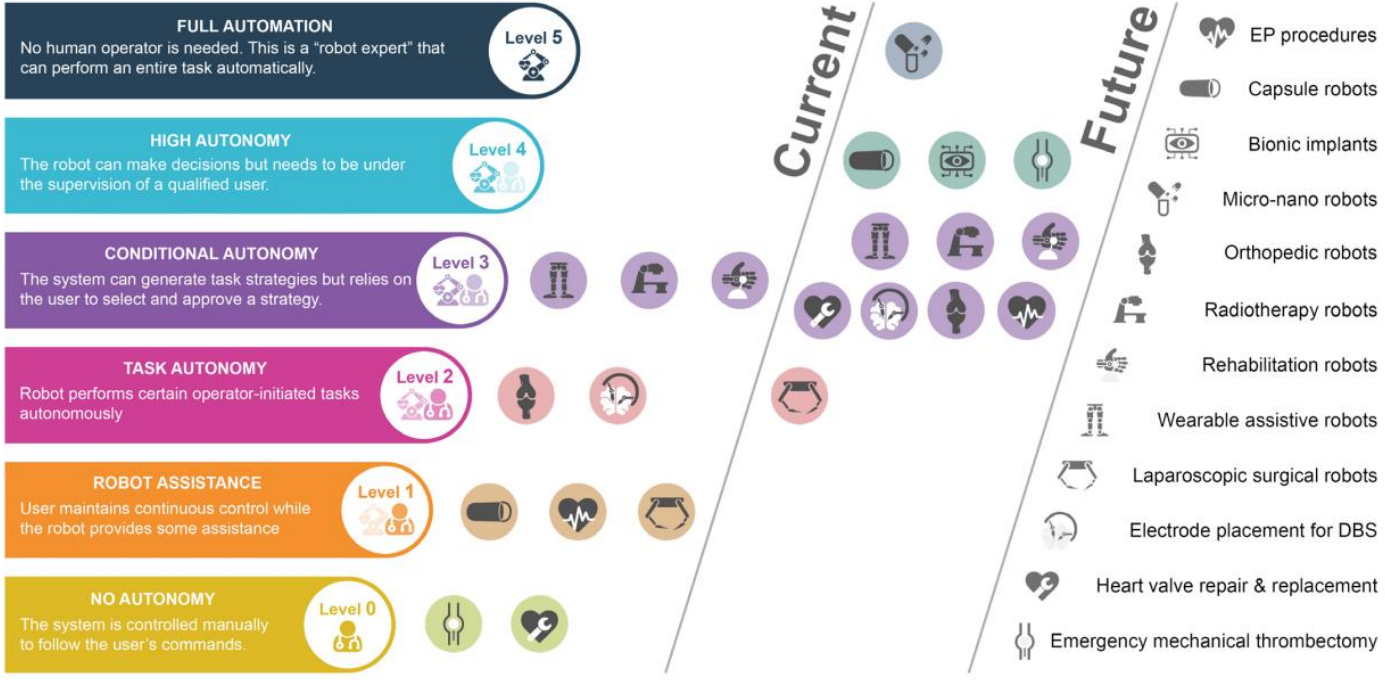
Current challenges and hot topics in medical robotics



Pierre E. Dupont, Bradley J. Nelson, Michael Goldfarb, Blake Hannaford, Arianna Menciassi, Marcia K. O'Malley, Nabil Simaan, Pietro Valdastri, "A Decade Retrospective of Medical Robotics Research from 2010 to 2020", *Science Robotics*, vol. 6, no. 60, 2021

Medical Robotics Development and Trends

Which level do we want?

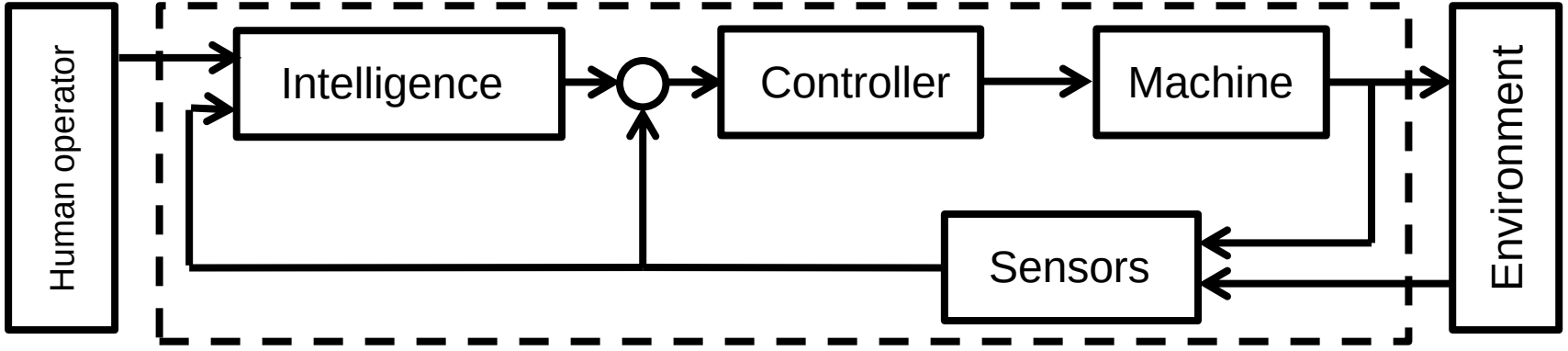


Pierre E. Dupont, Bradley J. Nelson, Michael Goldfarb, Blake Hannaford, Arianna Menciassi, Marcia K. O'Malley, Nabil Simaan, Pietro Valdastri, "A Decade Retrospective of Medical Robotics Research from 2010 to 2020", *Science Robotics*, vol. 6, no. 60, 2021

What Makes a Robotic System?

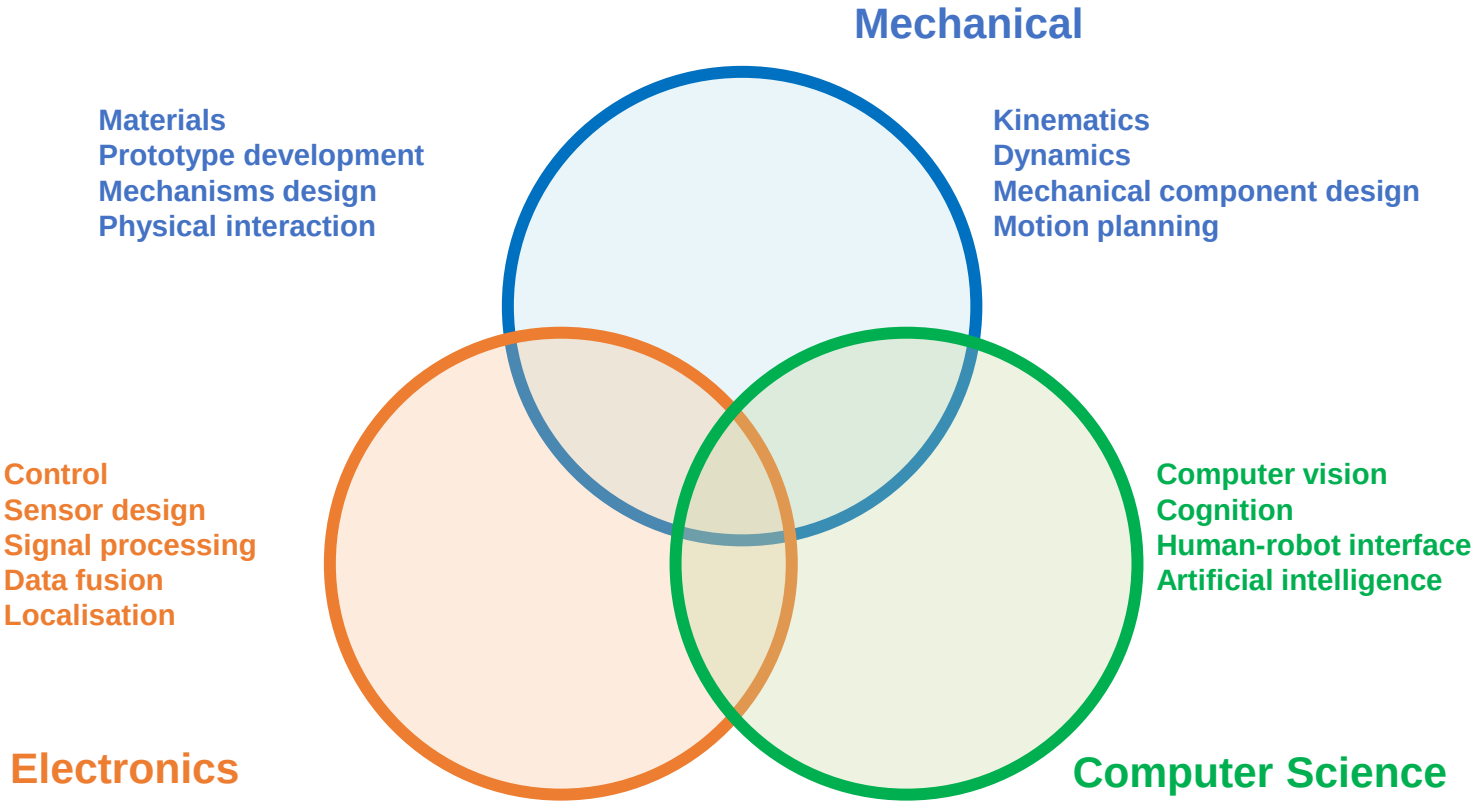
Common components of a robot system

- Plant (mechanics)
- Sensors (feedback)
- Controller (error correction)
- Intelligence (high-level decisions)
- Human operator (human-robot interaction)

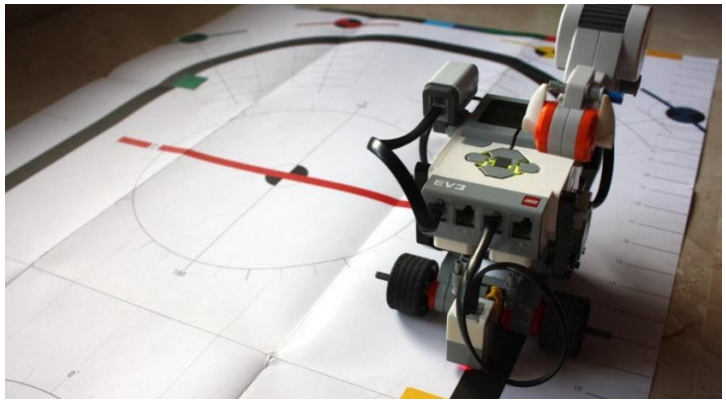


What Makes a Robotic System?

Robotics is a very broad field



The Development of Robots



Robots within the classroom



Early stage robots within labs

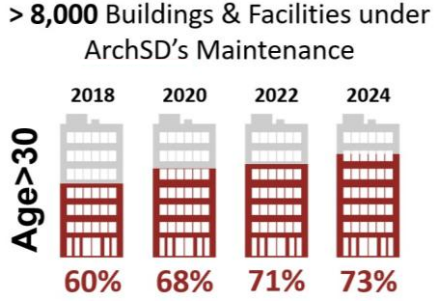


Robot testing in the field



Robots in the real-world

Façade Inspection Motivation



Façade inspection for spalling is important for public safety

- Contact based inspection (not only visual)
- Currently performed by humans on scaffolding and gondolas
- Over 6,000 public and 27,000 private buildings older than 30 years in Hong Kong





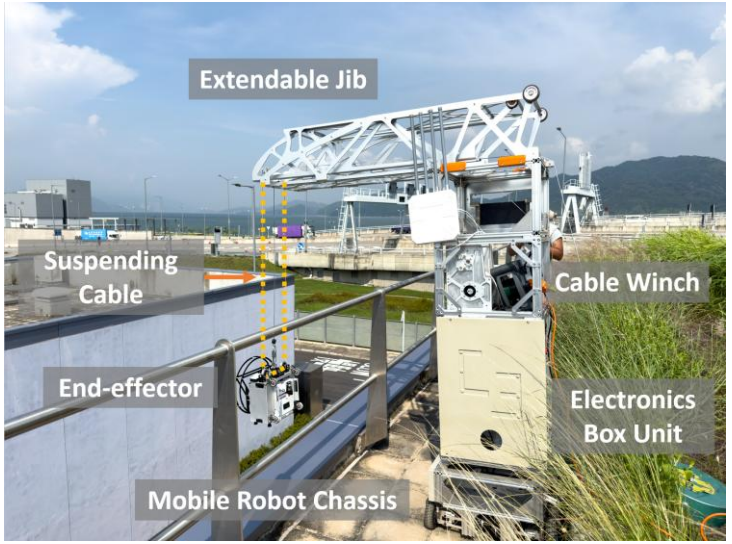




Comparison with Existing Systems

Our robot system:

- Single suspended cable with thruster stabilisation
- Robot-base with high mobility
- Physical hammer tapping
- Extension for non-flat surfaces





Significant features:

- **Lightweight** system that is **easy to setup**
- **No fixture required** to mount on the building
- Allow for operation on **non-flat façade surfaces**
- **More objective** and **accurate** over vision and IR-based inspection

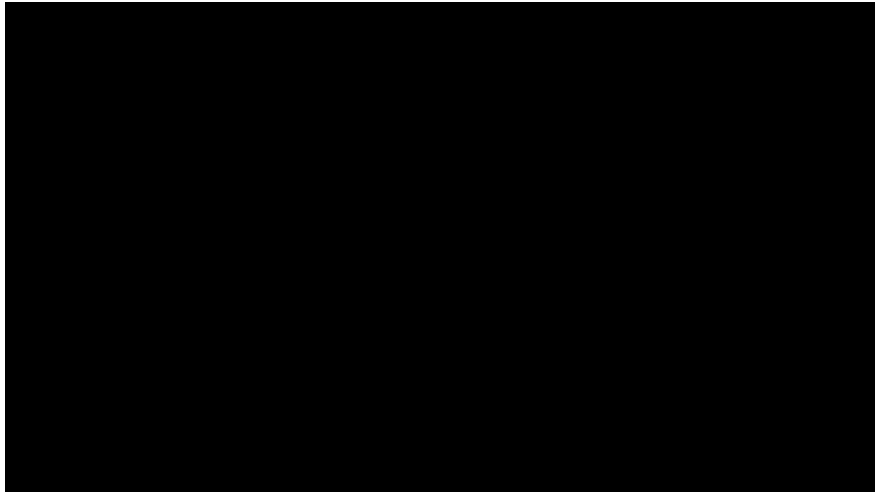
Robot Operation at Site Trial

Benefits to the workers and contractors

- **Increase work safety** of inspection workers, **lower time and financial costs** for inspection

Benefits to the public

- Improve inspection reliability and maintaining **higher quality living environment**





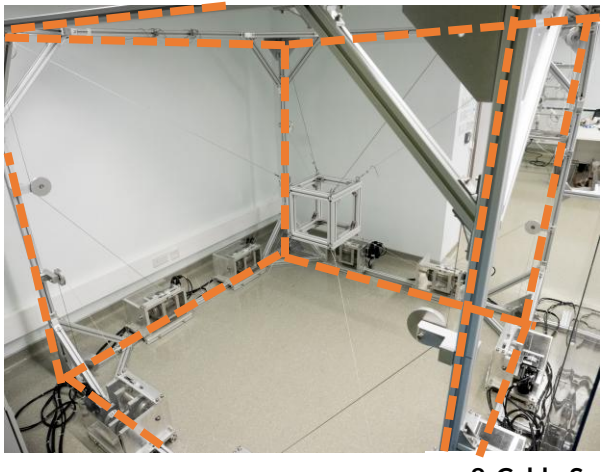
New Type of Cable-Driven Robot

Existing cable-driven robots:

- Require either 8, 4, or at least 2 cables
- Require a fixed mounting structure



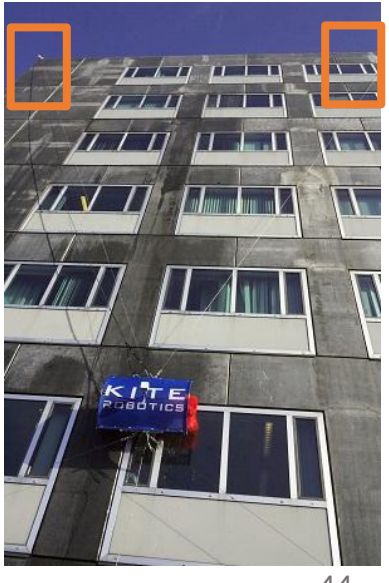
Our Mobile Cable Robot





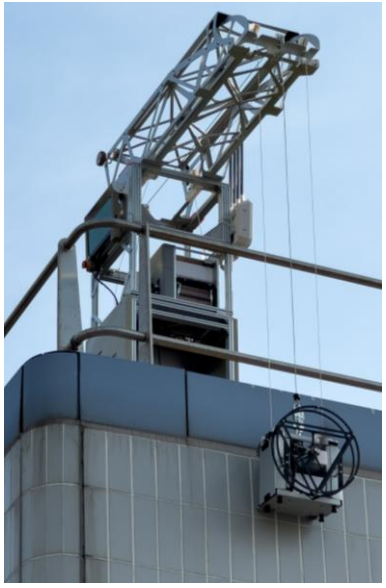
8-Cable Spatial Cable Robots

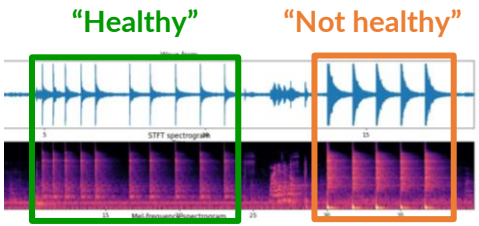




4-Cable Façade Cleaning Robots

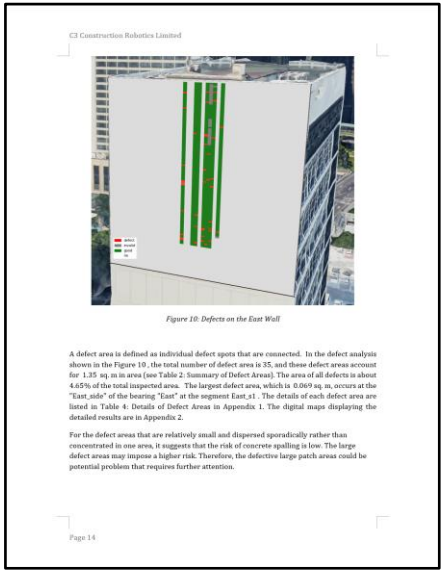
Inspection Workflow











RoboTapper | Cloud Server | Results | Report

Development Journey

2022



Q4
Collaboration with HK Architectural Services Department (ArchSD)

Prototype of RoBosun-Tapper



2023



Q3
Completion of 8 Pilot Projects with HK ArchSD

RoBosun-Tapper v1.0



2024

CITF

Q2
RoBosun-Tapper entered CITF Pre-approved lists

Q4
Completion of 20+ Building Inspection with HK ArchSD, Housing Department, Hospital Authority, MTR

RoBosun-Tapper v2.0



2025



Q1
Completion of Site Inspection on electricity station of Hong Kong Electric

RoBosun-Tapper v3.0



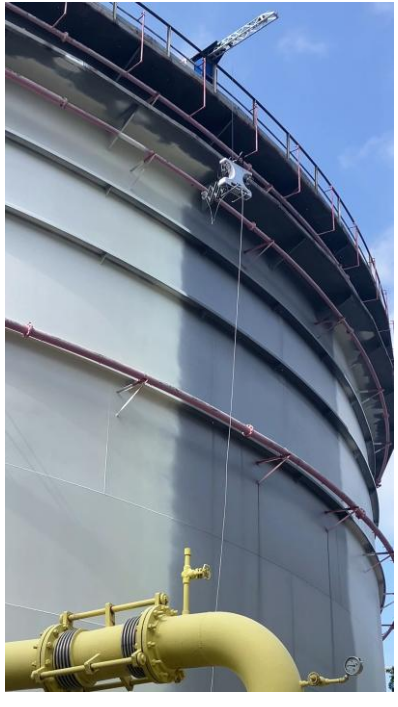
Real World Inspection



A Lightweight Painting Robot



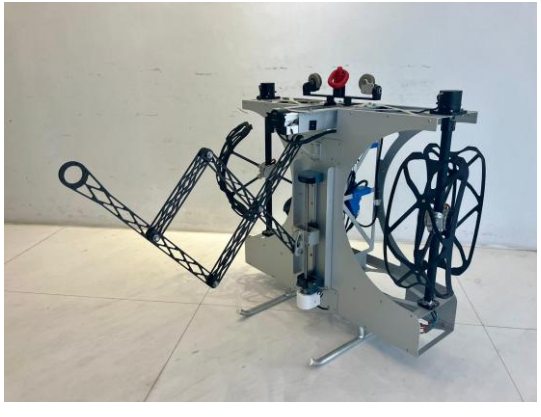
Testing on top of tank



Testing of painting on metal surface





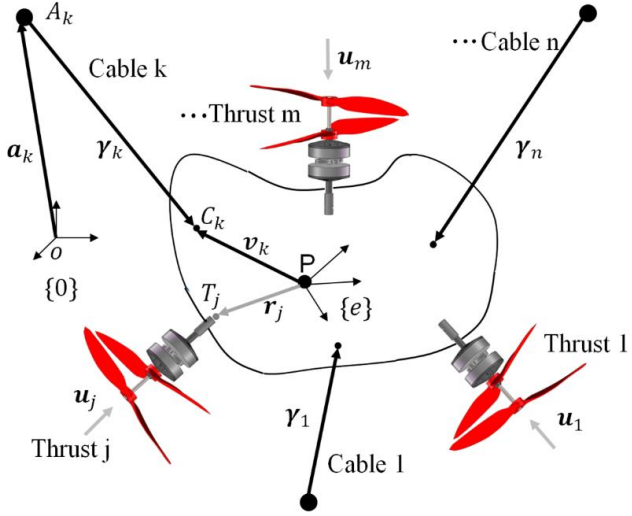


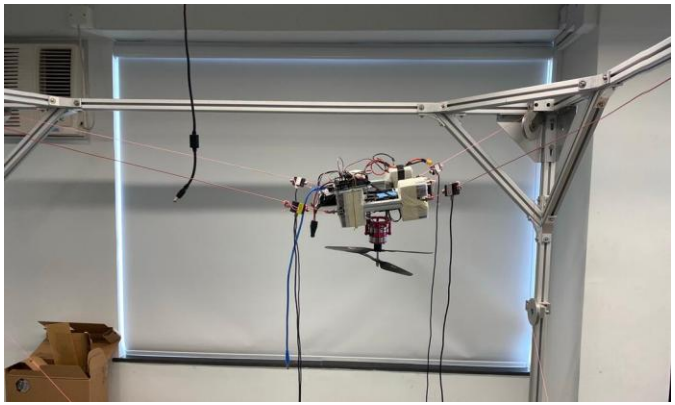
End-effector for painting

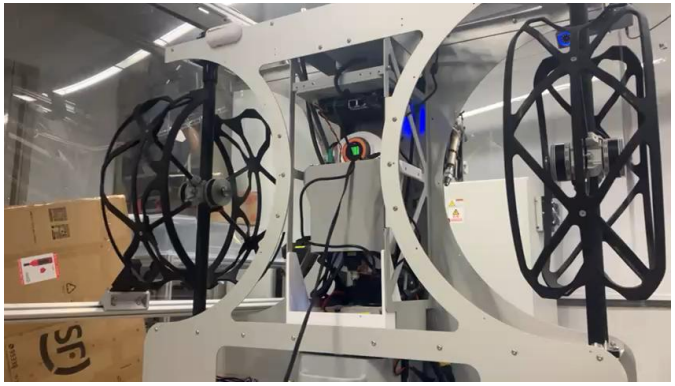
Thruster-Driven CDPRs (T-CDPRs)

Combine the advantages of CDPRs and UAVs

- CDPRs have high payload and long operation time
- Thrusters do not have environment interference



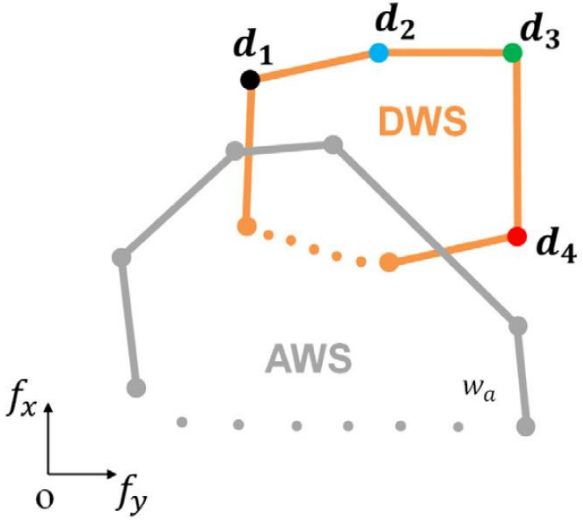


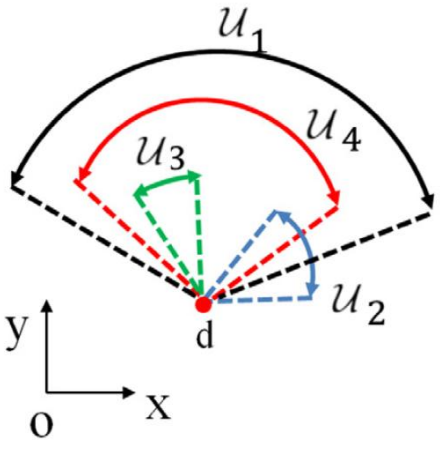


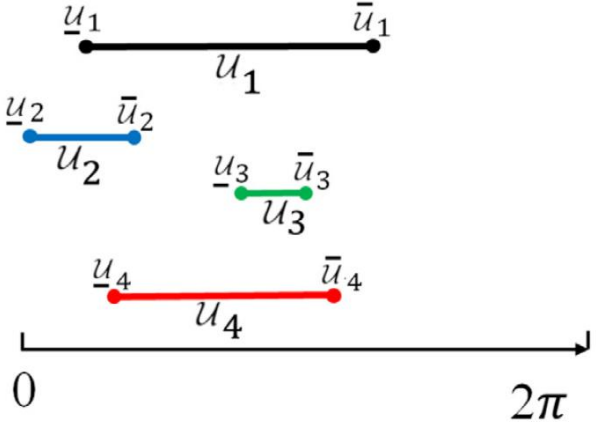
Y. Sun, Y. Guo, C. Song, and D. Lau, "Wrench-Feasible Workspace-Based Design of Hybrid Thruster and Cable Driven Parallel Robots", *Mechanisms and Machine Theory*, Vol. 172, pp. 104758, 2022

Thruster-Driven CDPRs (T-CDPRs)

Generate a set of feasible regions for infeasible wrenches



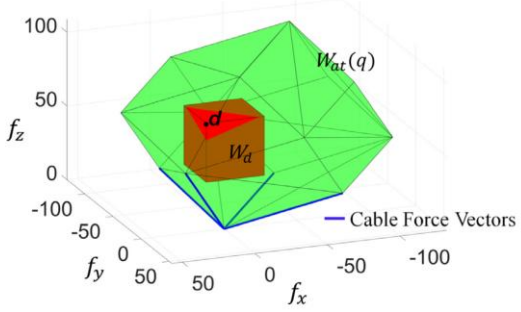


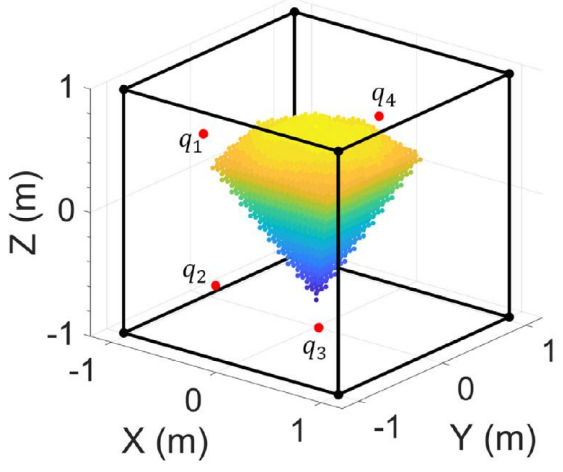


Y. Sun, Y. Guo, C. Song, and D. Lau, "Wrench-Feasible Workspace-Based Design of Hybrid Thruster and Cable Driven Parallel Robots", *Mechanisms and Machine Theory*, Vol. 172, pp. 104758, 2022

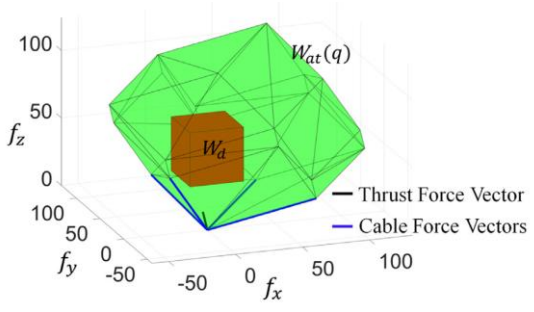
Thruster-Driven CDPRs (T-CDPRs)

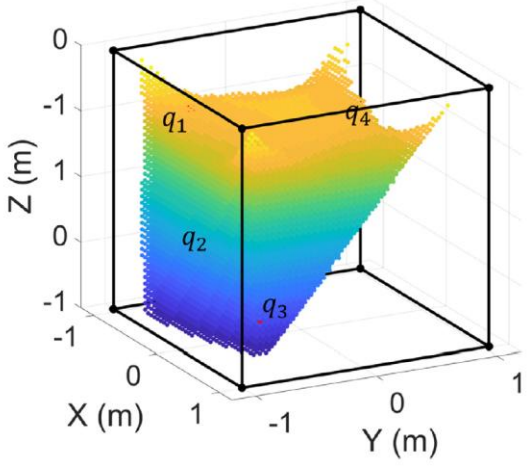
CDPR only





T-CDPR





Y. Sun, Y. Guo, C. Song, and D. Lau, "Wrench-Feasible Workspace-Based Design of Hybrid Thruster and Cable Driven Parallel Robots", *Mechanisms and Machine Theory*, Vol. 172, pp. 104758, 2022

Form and Function

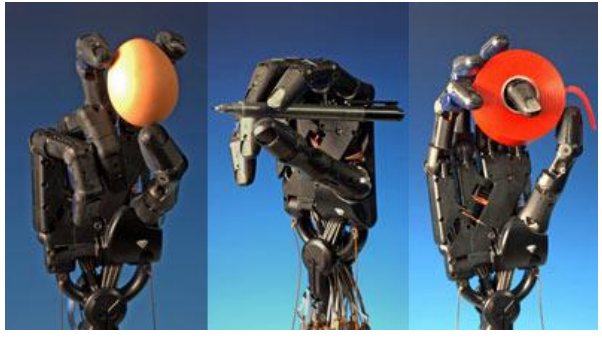
Weeks 1 and 2: Robot Manipulators

Robot Arms and Hands

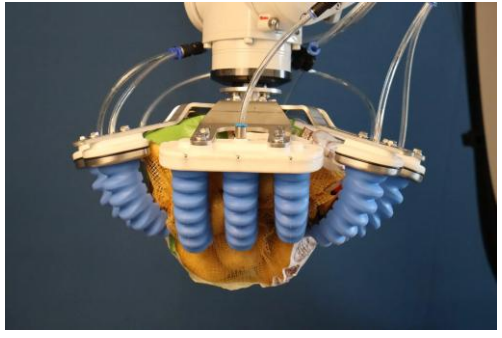
Goal to grasp and manipulate objects



Kuka LWR (KUKA Lightweight robot)



Shadow Robotics hand



Soft gripper



Righthand robotics Reflex hand



Universal Robotics UR3



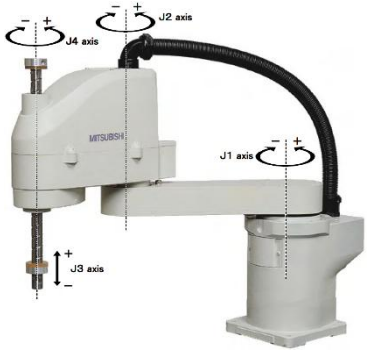
Superlimbs

Robot Arms and Hands



Robot Arms and Hands

[redacted]







[redacted]

Robot Arms and Hands

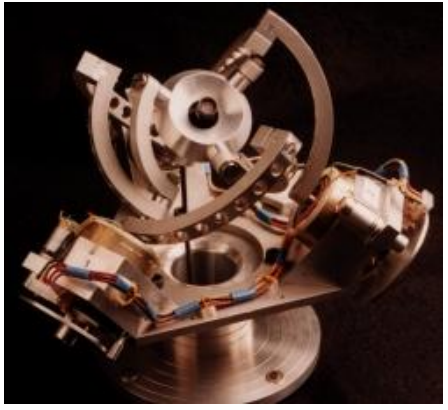


Parallel Robots









Top left: Lufthansa using Stewart platform
Top right: ABB FlexPicker delta robot
Bottom left: Fraunhofer IPA cable simulator
Bottom right: Laval University Agile Eye

Parallel Robots



Parallel Robots

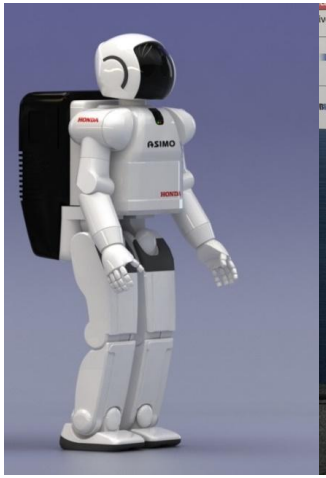




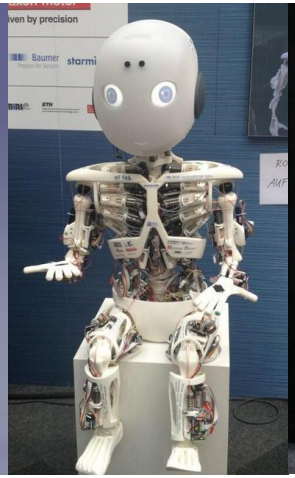


Week 3: Humanoid Robots

Humanoid Robots



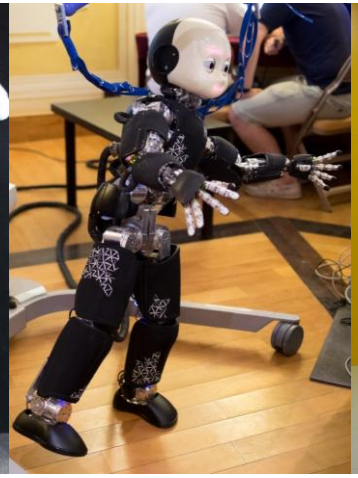
Honda ASIMO
(Japan)



Devanthro Roboy
(Switzerland)



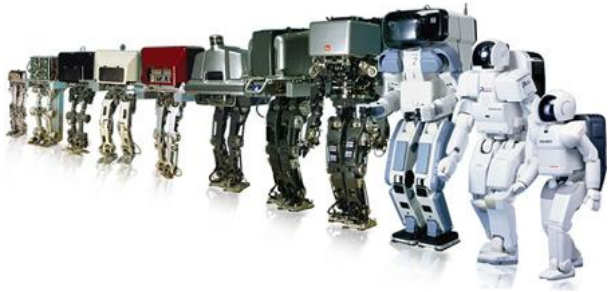
JSK Kojiro
(Japan)



RobotCub iCub
(EU)



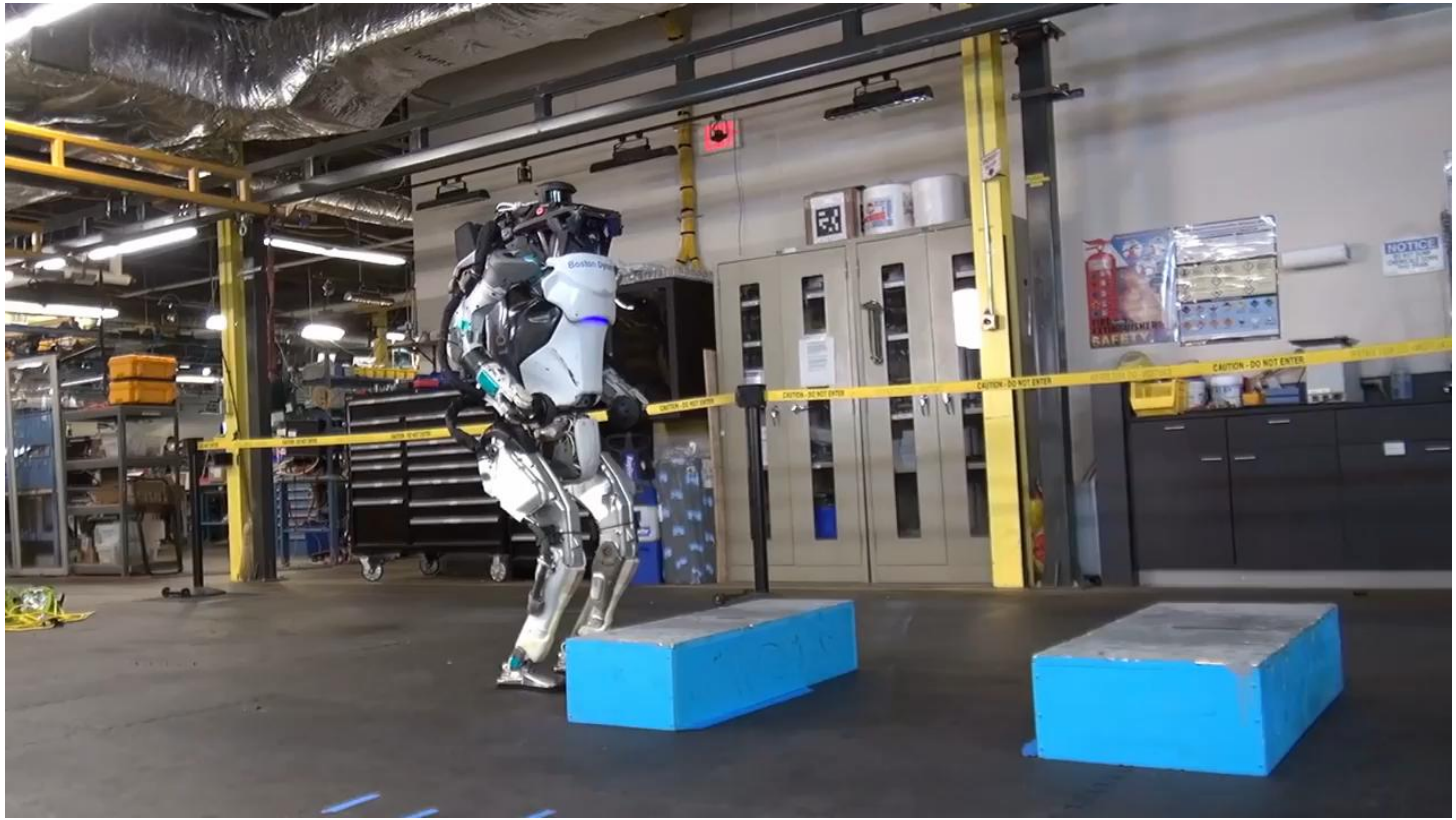
AIST HRP-4
(Japan)



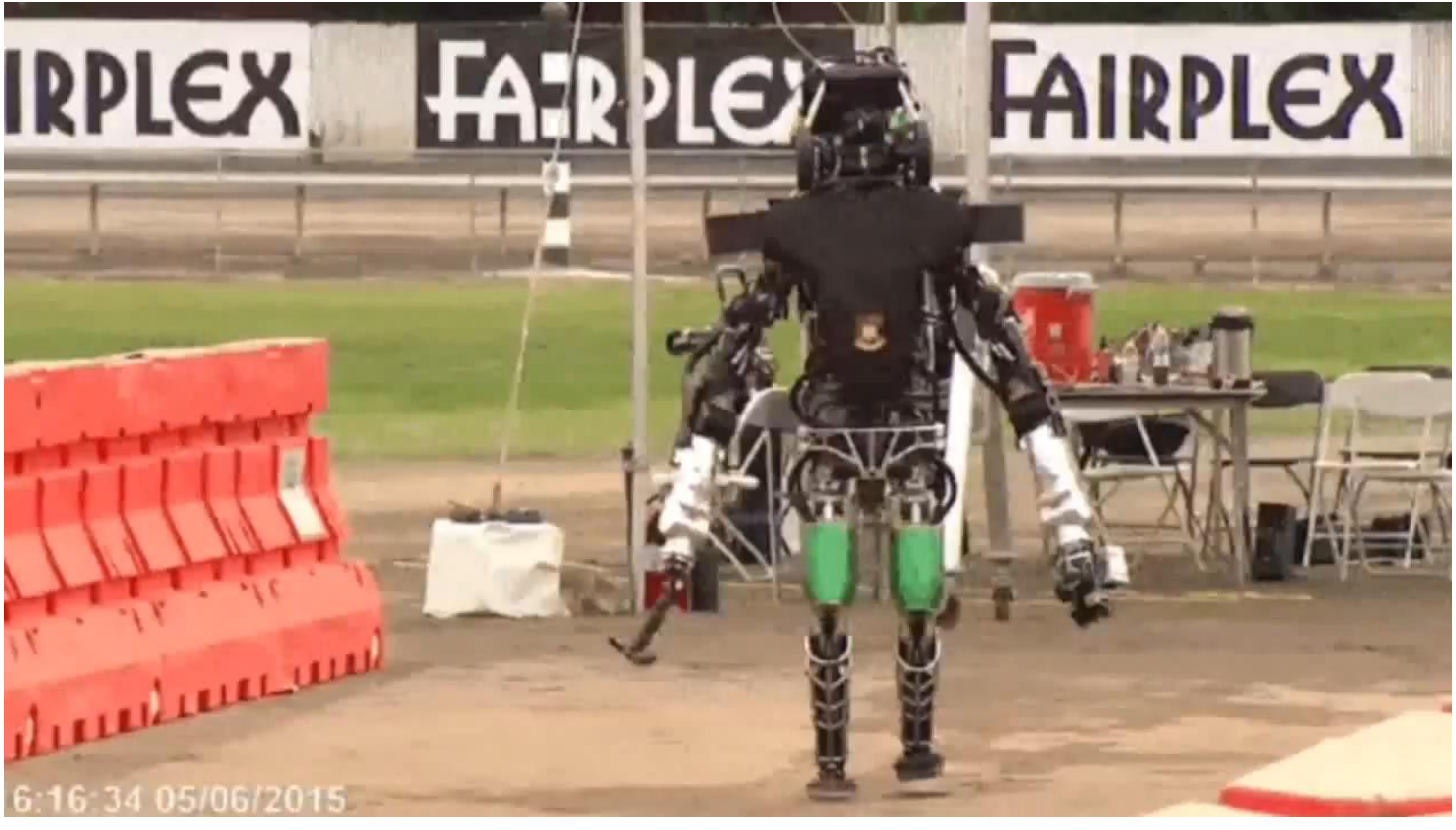
Humanoid Robots



Humanoid Robots

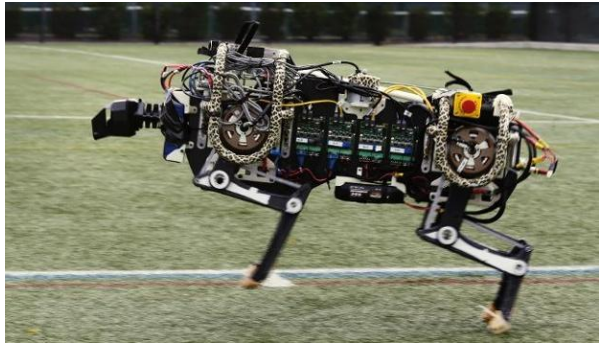


Humanoid Robots

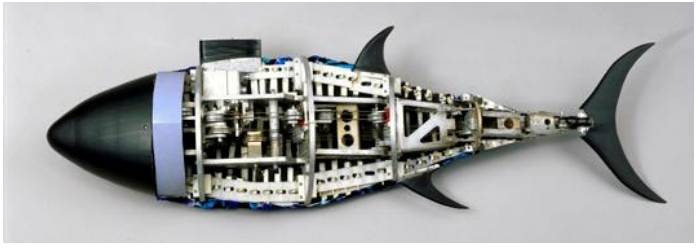


Week 4: Bio-Inspired Robots

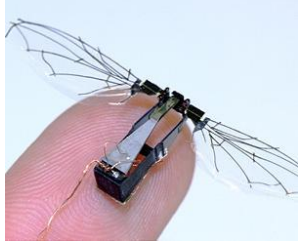
Animal-inspired Robots



MIT Cheetah robot



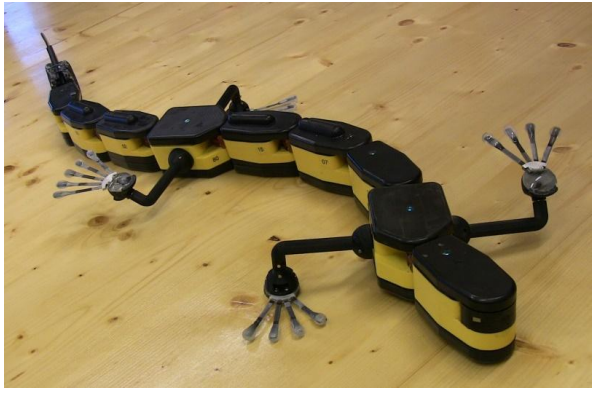
RoboTuna



Harvard Microfly



CMU snake robot



EPFL salamander robot

Animal-inspired Robots



Animal-inspired Robots



Week 5: Mobile Robots

Mobile Robots



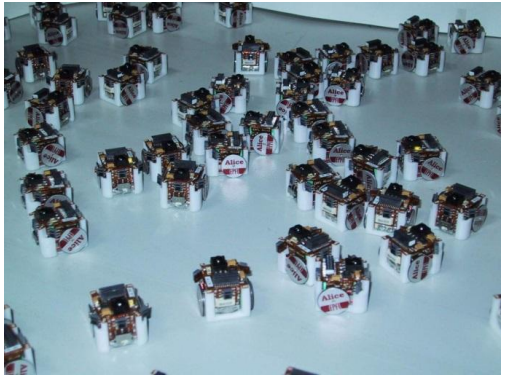
Aldebaran Robotics and Softbank mobile's Pepper service robot



iRobot PackBot



TU Eindhoven soccer robots



EPFL Alice swarm robot

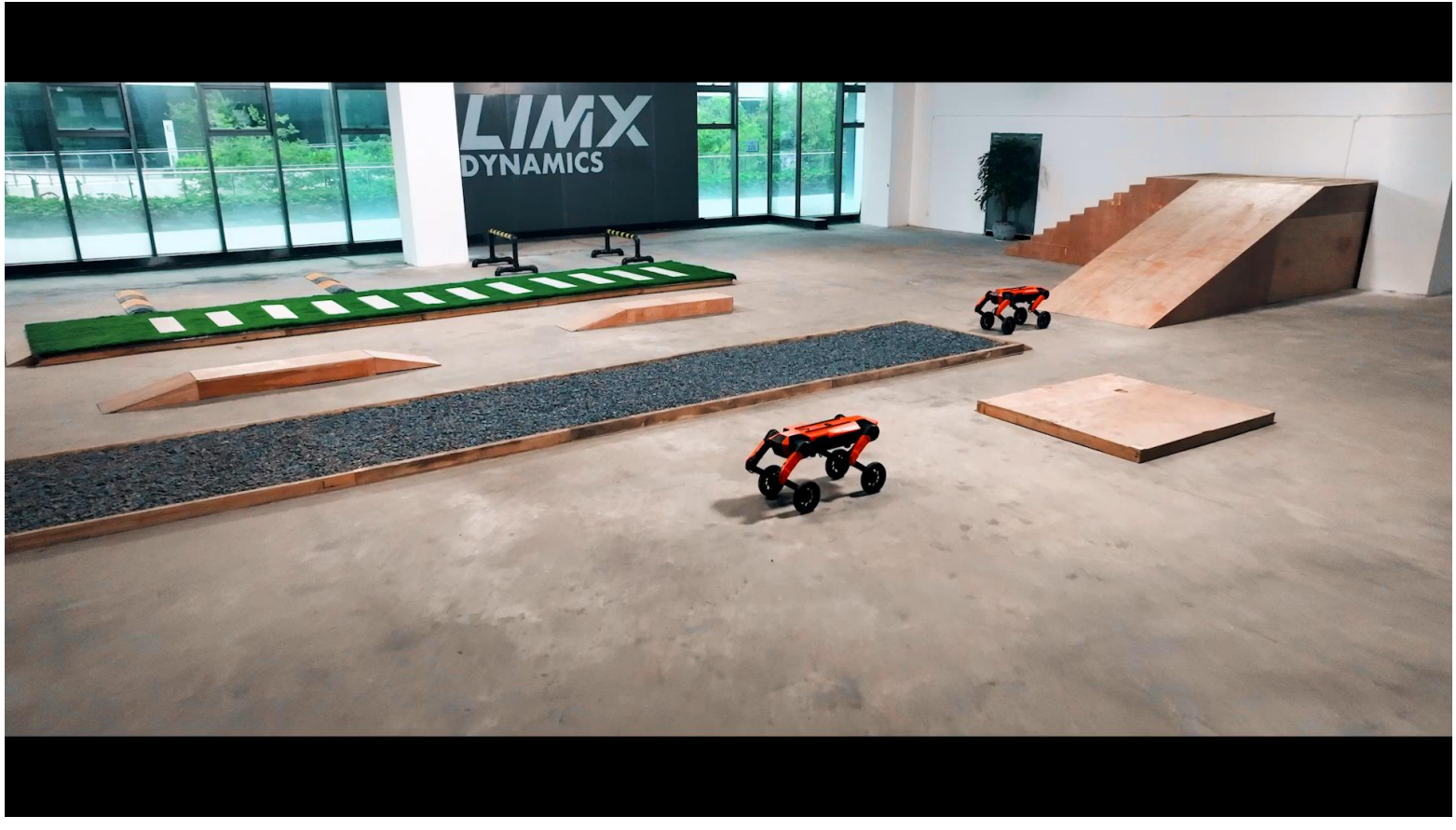


Cambridge Unmanned Aerial Vehicle

Mobile Robots: Mobile Manipulator



Mobile Robots: Wheeled Legged Robots



Mobile Robots: Flying Swarms



Mobile Robots: Slime Robots



Week 6: Manufacturing

Industrial Robots

Combine the advantages of CDPRs and UAVs

- Improve productivity: fast speed, low maintenance, long working hours (24/7)
- Improve quality: high accuracy and high repeatability

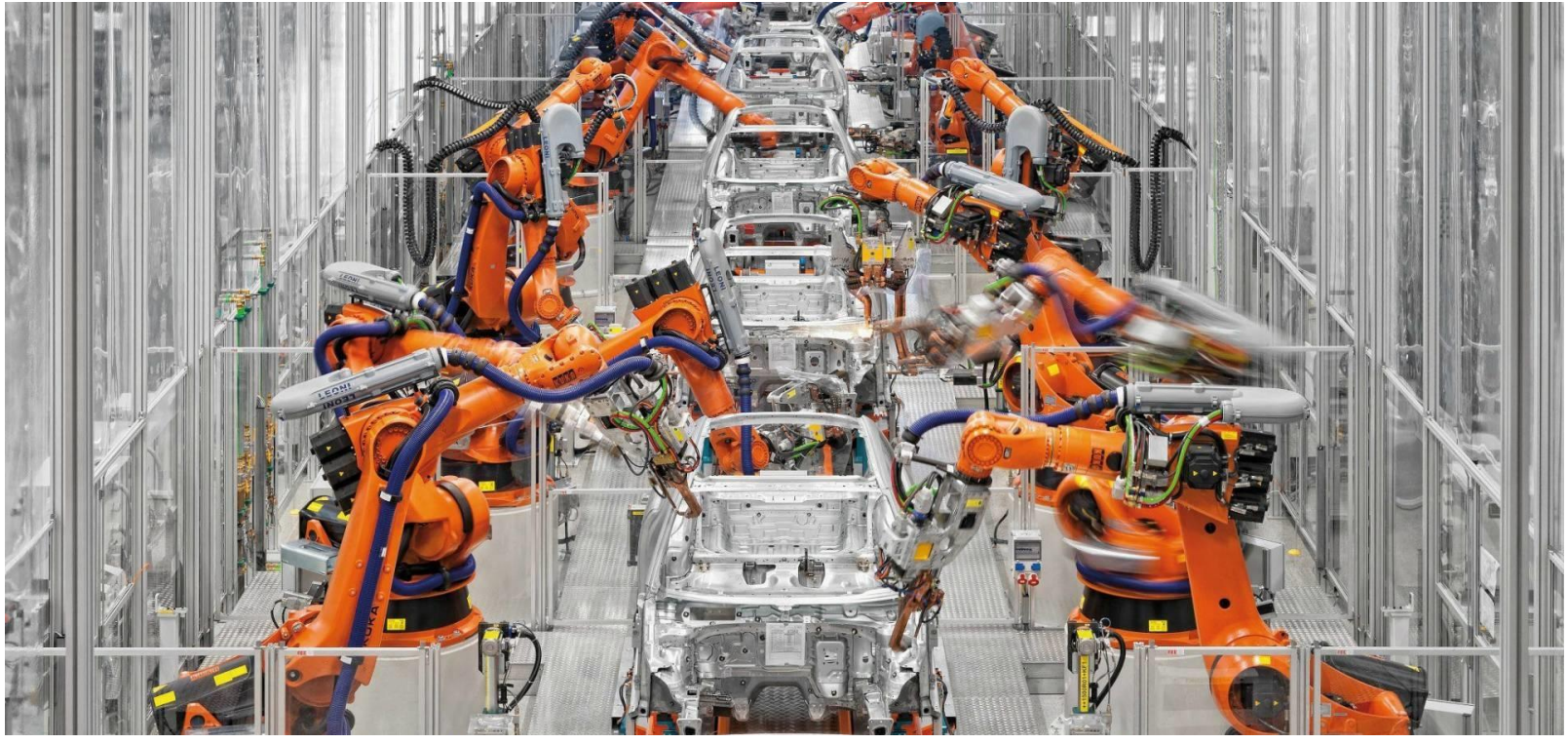
Timeline

- Early 1970s: Industries started using robots
- Late 1970s and early 1980s: Robots widely used in automobile and electronic industries

Applications are extremely widespread now

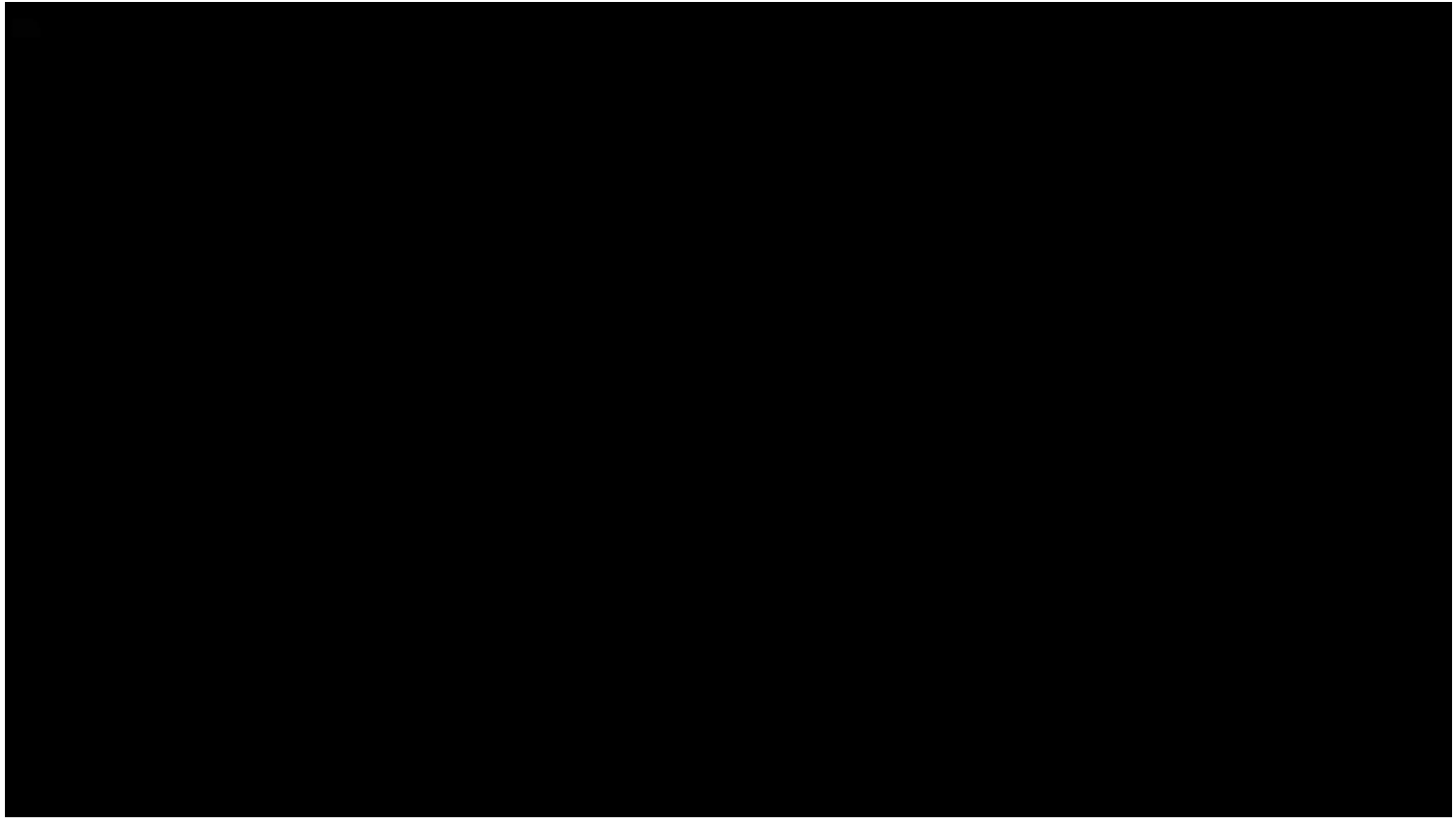
- Any repetitive tasks: painting, assembly, welding, cutting
- Tasks that could be automated: vision inspection, monitoring

Industrial Robots: Car Factory



KUKA robots working in a car factory

Industrial Robots: Pancake Handling



Industrial Robots: Car Production



Week 7: Agriculture Robots

香港中文大學
The Chinese University of Hong Kong

Agriculture Robots

Agribotics help automate tasks at farms

- Shoot weeds, monitor and tend crops

Challenges in environment sensing

- How to localise and manoeuvre within such a large space
- How to sense at the farms and react, vision, soil sensors etc.



Australian Centre of Field Robotics: RIPPA

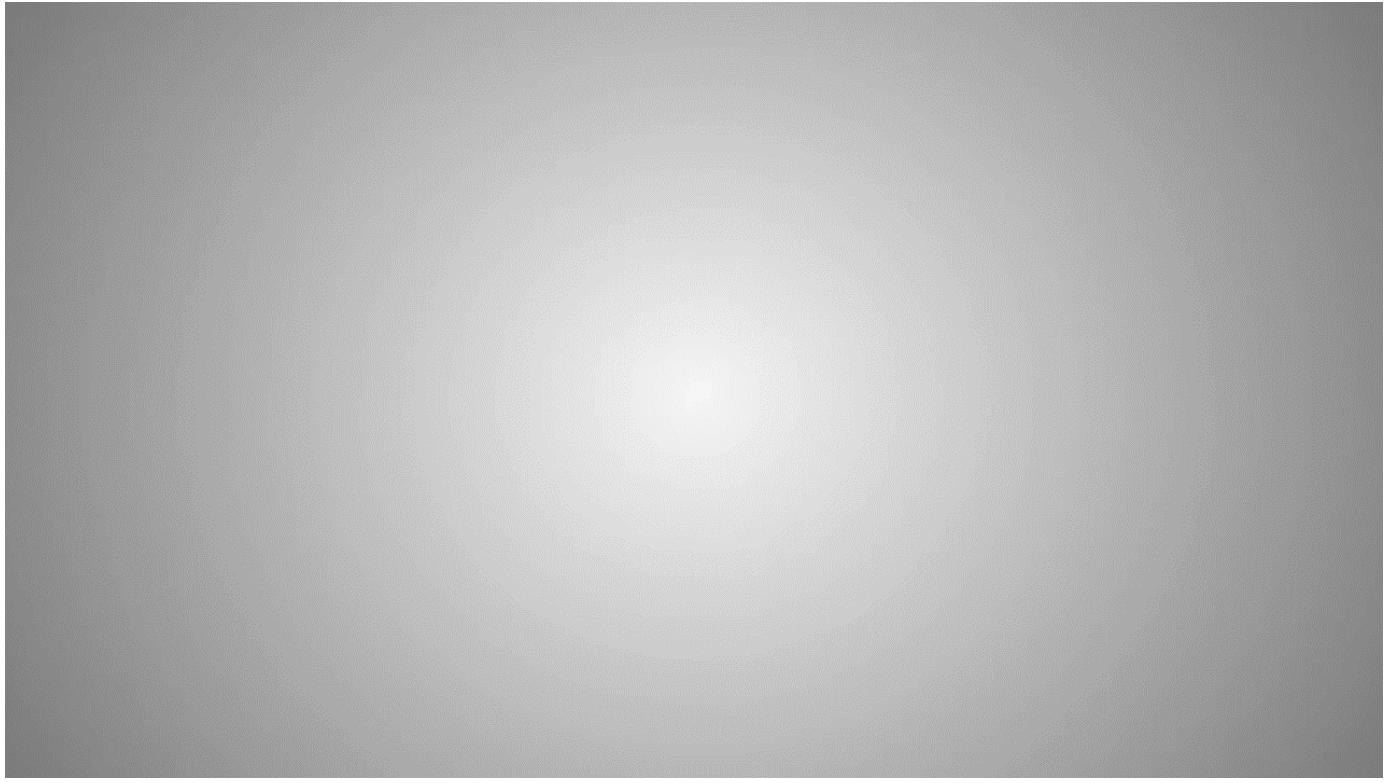


Universidad Politécnica de Madrid (UPM): Rosphere



UAV at vineyards

Agriculture Robots

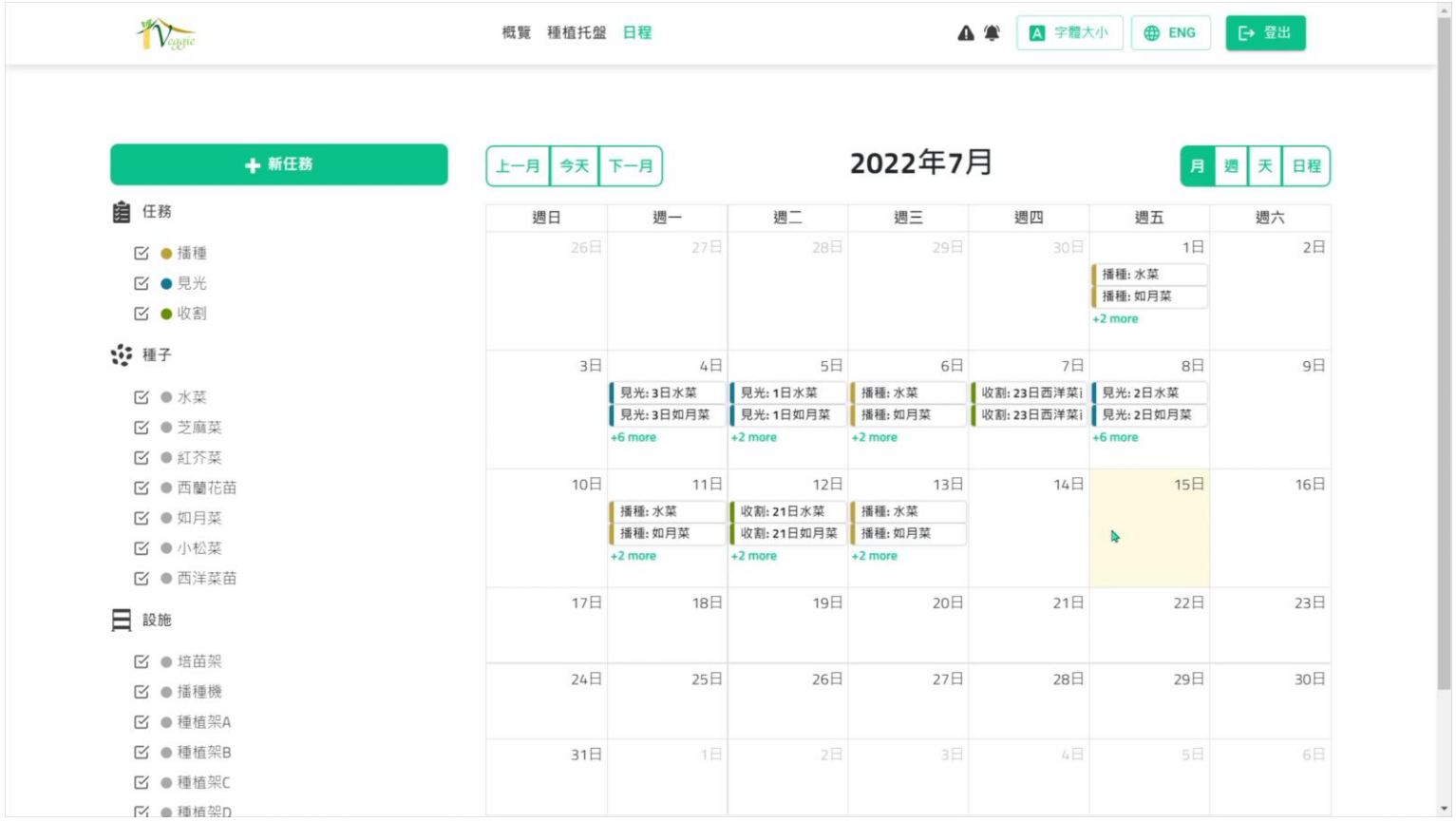


Agriculture Robots



香港中文大學
The Chinese University of Hong Kong

Agriculture Robots



Week 7: Extreme Environment Robots

Hazardous Environments

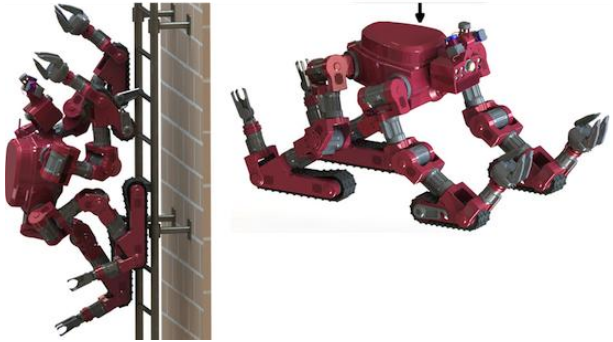
Take advantage of robots to operate in environments that humans cannot operate in or are dangerous to

- Small environments
- Large spaces
- Extreme hazardous environments

Some challenges include:

- How to design robots to operate in such unstructured environments
- How to localise effectively in the environment
- How to operate as a team to improve search efficiency
- How to manipulate and interact with the environment

Hazardous Environments

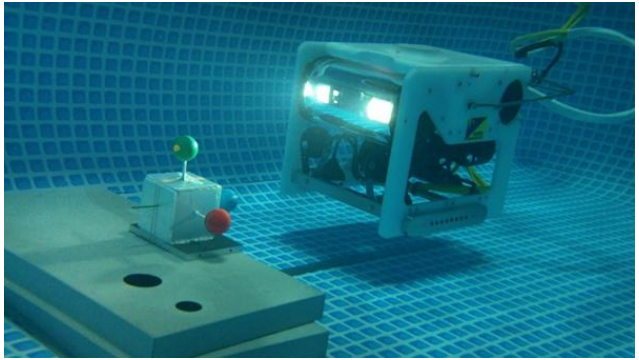


CMU CHIMP humanoid-tank robot





CMU Biorobotics lab and Ryerson University's NCART lab : rescue dog and snake robot



Okayama University Autonomous Underwater Vehicle

Hazardous Environments: Ocean-1



Week 8: Entertainment Robots

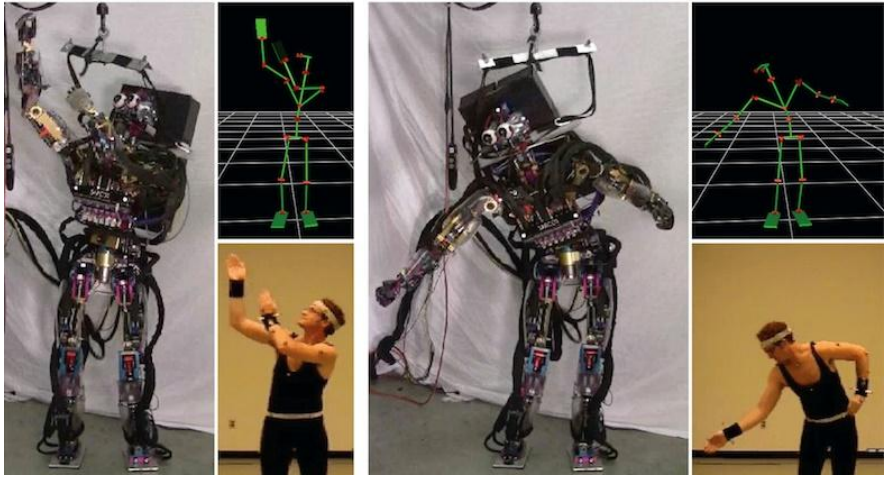
Entertainment Robots

Promising area that everyone enjoys

- For example, Disney research laboratories and LEGO



Music playing robots by Toyota

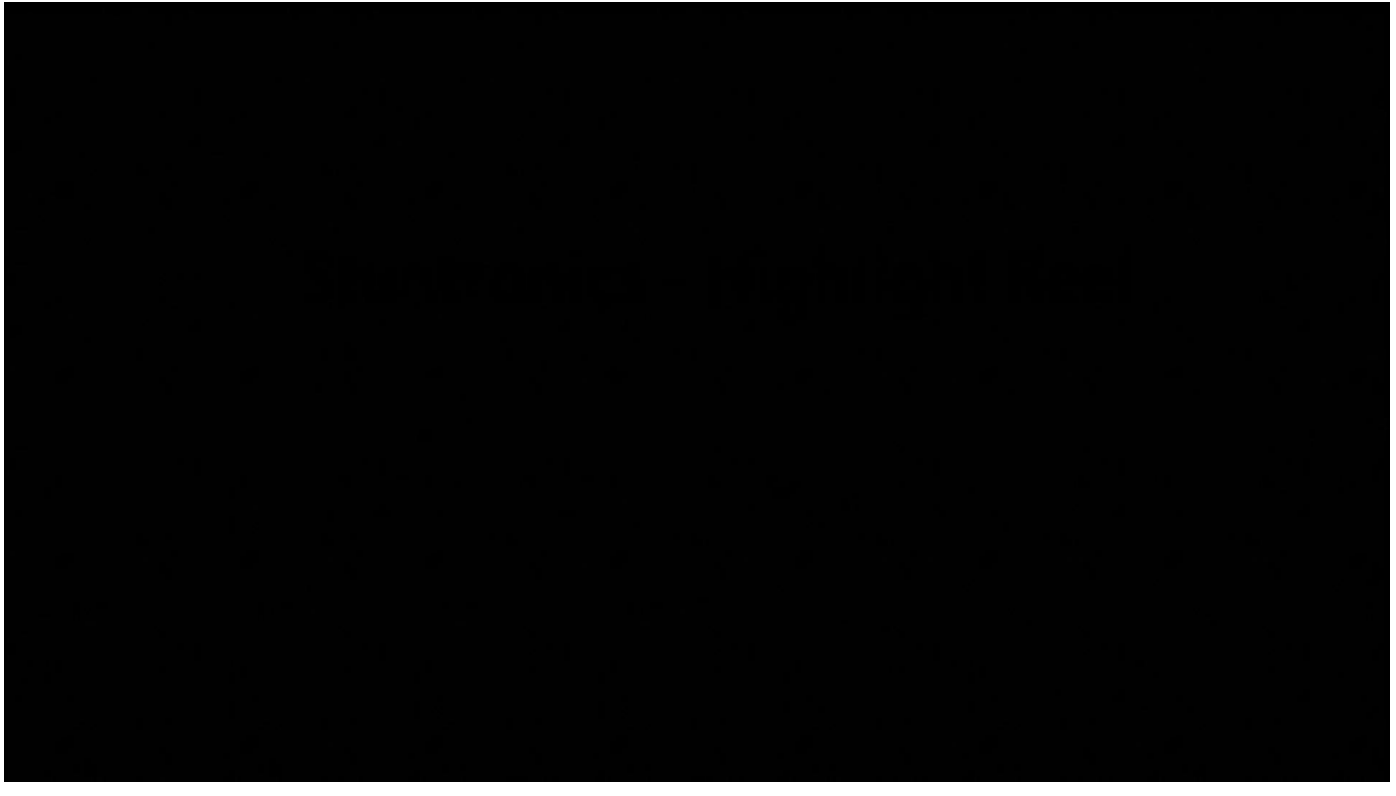


Disney Research: human motion replication using motion capture

Entertainment Robots: Soccer Playing



Entertainment Robots: Stunt Robot



Entertainment Robots: Drone Fireworks

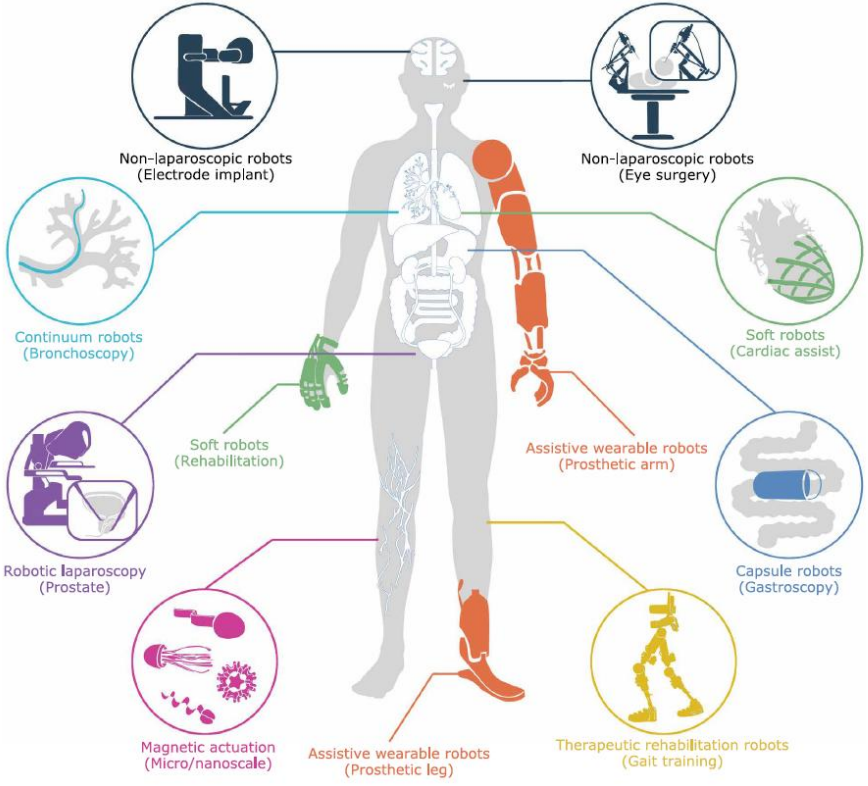


Entertainment Robots: Disney Character Robot



Week 9: Medical Robots

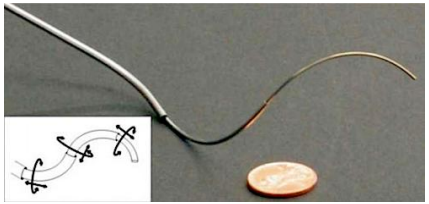
Medical Robotics



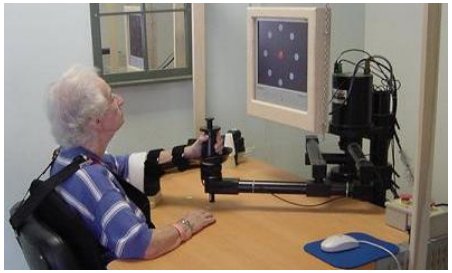
Medical Robotics

An area that has great impact to the society

- Rehabilitation robotics and prosthetics
- Surgical robotics
- Micro/nano manipulation for cell-manipulation



Vanderbilt University: steerable



MIT Newman Laboratory for Biomechanics and Human Rehabilitation



MIT Media Lab: bionic ankle-foot



ETHZ: Nanomag on an inverted microscope for cell manipulation

Medical Robotics: Prosthetics



Medical Robotics: Surgical Robot



Medical Robotics: Magnet Surgical Robot



Week 10: Entertainment Robots

Service Robots

Another very promising field that fits the society's needs

- Aging population increasing
- Robots will become more common in the society



Aldebaran Robotics and Softbank mobile's Pepper service robot



Savioke's Dash robot at Crowne Plaza San Jose-Silicon Valley hotel

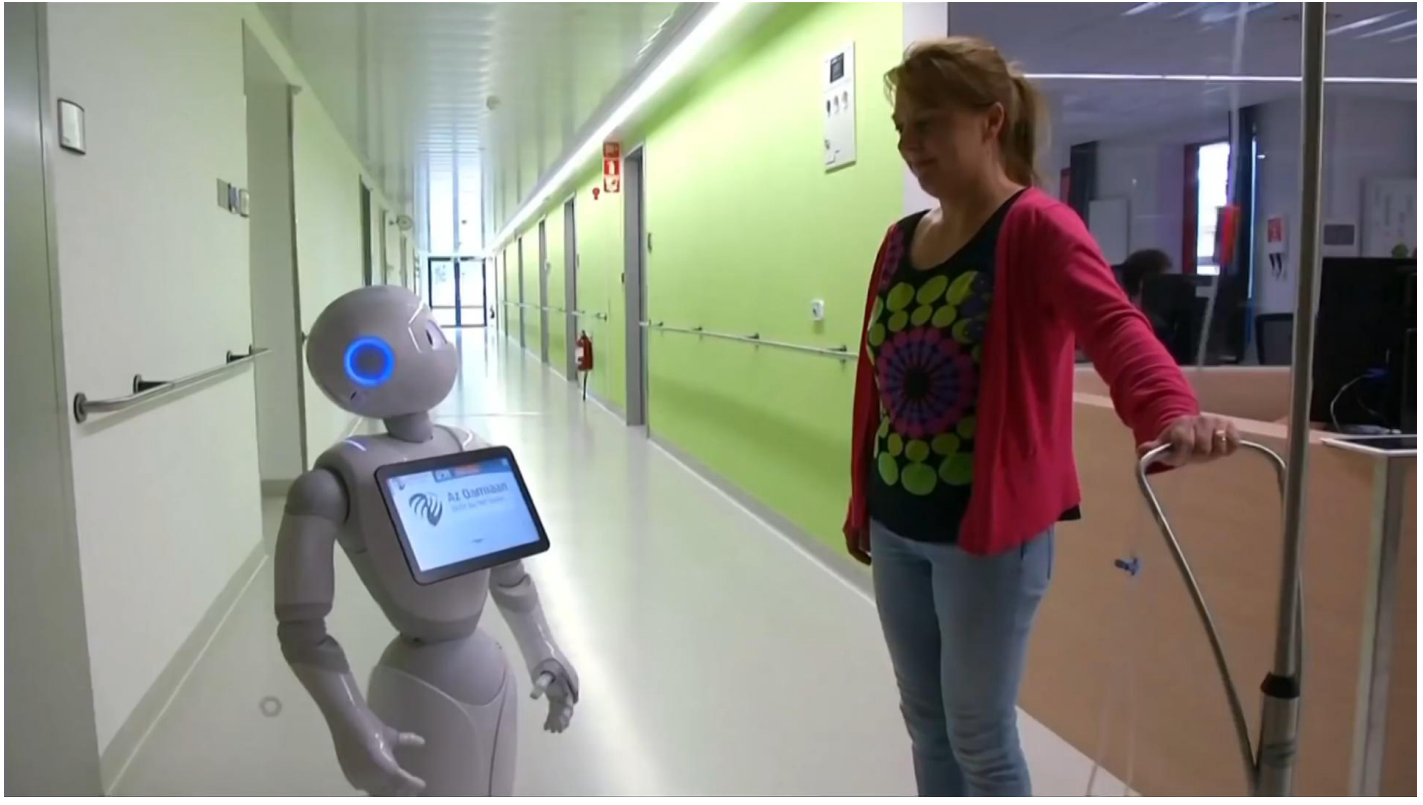


Fraunhofer IPA: Care-O-bot 4

Service Robots: Amazon Warehouse



Service Robots: Hospital Receptionist



Service Robots: Cooking Robot

